

Full Length Article

Long-short-term decoupled residual learning for high-resolution spatio-temporal air quality inference

Hui Zheng^{a,b}, Jialing Huang^{a,c}, Huiming Chen^{a,b}, Zhenlong Xiao^{a,b,*}, Nengwang Chen^d, Xinghao Ding^{a,b}

^a Key Laboratory of Multimedia Trusted Perception and Efficient Computing, Ministry of Education of China, Xiamen University, Xiamen, 361005, China

^b School of Informatics, Xiamen University, Xiamen, 361102, China

^c Institute of Artificial Intelligence, Xiamen University, Xiamen, 361102, China

^d Fujian Provincial Key Laboratory for Coastal Ecology and Environmental Studies, College of the Environment and Ecology, Xiamen University, Xiamen, 361102, China

ARTICLE INFO

Keywords:

Air quality inference
Multi-source data fusion
Static and dynamic features fusion
Residual correction

ABSTRACT

Accurate high-resolution spatio-temporal air quality inference based on sparse monitoring stations is essential for environmental governance and public health. However, prevailing deep learning approaches often treat inference as direct regression on observed air quality concentrations, neglecting the distinct effects of a long-term static baseline and short-term dynamic perturbations on the inference. This limits the model's generalization ability and may cause spatial bias in the inference. In this paper, we propose a novel Spatio-Temporal Air Quality Inference Network (SAQIN) model, which provides an explicit decomposition of the static and dynamic effects in the inference. The static branch integrates high-resolution semantic segmentation, global context from remote sensing, population density, and elevation to construct a high-fidelity environmental prior. The dynamic branch employs multi-head self-attention mechanism to jointly encode geographic relationships and inter-pollutant chemical coupling effects. We further develop a residual learning module to model the inference process as a residual correction anchored to multiple reference stations and aggregates predictions through stepwise multi-station fusion with distance-weighted averaging, which successfully eliminates the Voronoi-partition-induced discontinuities and yielding a globally smooth physically plausible air quality inference field. SAQIN has been shown to demonstrate superior accuracy and strong zero-shot cross-domain generalisation when evaluated on large-scale real-world datasets, thus outperforming state-of-the-art methods across a range of criteria pollutants.

1. Introduction

Air pollution poses a severe threat to public health, ecological sustainability and climate change (Fadhel et al., 2024; Huang et al., 2021; Organization et al., 2015). Accurate and high-resolution spatio-temporal inference of air quality is therefore critical for environmental governance and public health interventions. Although ground-based monitoring stations are able to provide reliable measurements, their deployment remains sparse and uneven due to economic and logistical constraints (Huang et al., 2018). This limitation necessitates the employment of robust inference models capable of inferring continuous and physically plausible air quality fields from limited observations.

Existing research has addressed this challenge through two major paradigms. The first relies on classical spatial interpolation or regression techniques such as inverse distance weighting (IDW) (Wong et al., 2004), ordinary Kriging (Li & Heap, 2011), and land-use regression

(LUR) (Jutzeler et al., 2014; Lin et al., 2017; Patel et al., 2022; Wang et al., 2019). These methods often assume spatial stationarity or linear relationships, which leads to substantial performance degradation in the presence of strong nonlinearity and heterogeneity in urban air pollution. The second paradigm leverages machine learning and deep learning models, including Gaussian processes (Hu et al., 2023a), graph neural networks (Bonet et al., 2022), spatio-temporal field neural networks (STFNN) (Feng et al., 2024), and nationwide inference models such as AirRadar (Wang et al., 2025). While these approaches significantly enhance nonlinear modeling capacity, they remain constrained by the gap between the prevailing modeling paradigm and the physical law of atmospheric pollution fields.

Most methods regard air quality inference as direct regression of concentration values (Cheng et al., 2018; Han et al., 2023; Hu et al., 2023a; Song & Stettler, 2022; Su et al., 2024; Zhang et al., 2022a), overlooking that the observed concentrations arise from the superposition of two distinct components. One is a long-term static

* Corresponding author.

E-mail address: zxiao@xmu.edu.cn (Z. Xiao).

<https://doi.org/10.1016/j.neunet.2026.109575>

Received 23 November 2025; Received in revised form 25 March 2026; Accepted 29 August 2026

Available online 1 September 2026

0893-6080/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

baseline determined by stable environmental factors such as land use, population density, and terrain. This baseline reflects a region's inherent emission potential and dispersion capacity and exhibits pronounced spatial non-stationarity. The other component consists of short-term dynamic variations driven by transient atmospheric processes, including wind advection and photochemical reactions. These perturbations are physically constrained by atmospheric diffusion and therefore vary smoothly across space and time.

Regarding these two components as a single regression problem will cause the model to conflate distinct components mechanistically, degrading the performance of air quality inference. For instance, persistently high concentrations in industrial zones may be falsely attributed to transient meteorological conditions rather than ongoing emission sources. This conflation limits generalization in unseen regions and introduces systematic spatial bias into predictions. This misaligned paradigm has several interrelated shortcomings. For example, static and dynamic features with heterogeneous temporal scales are often combined with one another without making a clear distinction between them (Cheng et al., 2018; Song & Stettler, 2022; Zhang et al., 2022a). Remote sensing imagery is often underutilized—treated as generic pixel inputs or reduced to coarse indices such as NDVI (Su et al., 2024; Zaid & Basu, 2024), failing to explicitly represent pollution sources such as industrial zones and road networks. Moreover, inference strategies relying on a fixed set of local reference stations implicitly discretize the continuous atmospheric field into Voronoi-like cells (Bezyk et al., 2021), leading to concentration discontinuities at Voronoi-partition boundaries and violating the smoothness property of atmospheric diffusion.

To systematically address these issues, we propose the Spatio-Temporal Air Quality Inference Network (SAQIN), which reformulates the modeling paradigm from direct concentration regression to residual correction based on reference station observations. Specifically, air quality inference at specific locations are anchored to real measurements at reference stations, and the residual relationship between the targeted location and each reference station is investigated. This residual is then added to the observed reference value to reconstruct the final air quality inference results. The proposed SAQIN model can inherently remove the influence of the unknown regional static bias and focuses on modeling the physically smoothness of local diffusion field.

The proposed SAQIN explicitly decouples the static and dynamic effects in air quality inference. The static branch integrates high-resolution semantic segmentation and lightweight global context from remote sensing, which is further enhanced by population density and elevation, forming a high-fidelity environmental prior. This prior can distinguish between types of land use associated with pollution sources, such as industrial zones and roads. The dynamic branch uses multi-head self-attention to jointly model geographic relationships and inter-pollutant chemical couplings, adaptively capturing spatio-temporal variations. Thereafter, a residual correction mechanism has been developed to avoid introducing the unknown static global bias, ensuring robustness inference in data-sparse and cross-domain settings. Finally, a stepwise multi-station fusion module is designed to aggregate local predictions via distance-weighted averaging, eliminating Voronoi-induced discontinuities and yielding a smooth physically plausible inference field. We evaluate SAQIN on a large-scale dataset comprising 171 monitoring stations in the Yangtze River Delta. Experimental results demonstrate consistent performance improvement over state-of-the-art methods across the criteria air pollutants, with particularly strong zero-shot cross-domain transfer performance. The main contributions of this work are summarized as follows.

- SAQIN is developed as a dual-branch framework to explicitly decouple long-term static environmental priors from short-term dynamic variations in air quality inference. This distinguishes the essential differences in the effects of the inference problem at different temporal scales and enables physically interpretable inference results.

Table 1

Summary of key notations used in this paper.

Symbol	Description	
Location	$\mathbf{l}$	Geographic location (latitude, longitude)
	$\mathbf{l}_i$	The location of the i_{th} reference station
	$\mathcal{L}$	Set of all locations in the region of interest
	$\mathcal{L}_{obs}$	Set of monitored locations
Input	I_{RS}	Input remote sensing image
	ρ, h	Population density and Elevation
	d_i, θ_i	Distance and bearing to the i_{th} reference station
	$\mathbf{P}_T$	Sinusoidal time embedding vector
	C_i	Observed concentration at the i_{th} reference station
Output	$y(\mathbf{l}, t)$	Ground-truth pollutant concentration at location $\mathbf{l}$ and time t
	$\hat{y}(\mathbf{l}, t)$	Inferred pollutant concentration
	$\Delta \hat{y}_i(\mathbf{l}, t)$	Predicted concentration offset relative to the i_{th} reference station
	$\hat{y}_i(\mathbf{l}, t)$	Inferred concentration relative to the i_{th} reference station
Feature	$\mathcal{F}^{static}$	Static feature of target location
	$\mathcal{F}^{(i)}_{static}$	Static feature of the i_{th} reference station
	$\mathcal{F}^{dynamic}_{(i)}$	Dynamic feature of the i_{th} reference station
	$\mathcal{F}_{Seg}$	Semantic feature of I_{RS}
	$\mathcal{F}_{global}$	Global contextual feature of I_{RS}
	$\mathcal{F}_{RS}$	Fused representation of I_{RS}
	$\mathcal{F}_{tab}$	Tabular feature from ρ and h
Other	N	Number of selected reference stations
	α_i	Distance-based weight for the i_{th} reference station

- The static branch integrates high-resolution semantic segmentation of different types of land use with lightweight global context, population density and elevation data to provide accurate spatial non-stationary priors. The dynamic branch uses multi-head self-attention to jointly capture geographic proximity and inter-pollutant chemical couplings to adapt to short-term variations.
- A residual correction mechanism based on reference station observations together with step-wise multi-station weighted fusion module are designed to alleviate unknown static global bias and eliminate Voronoi-partition-induced discontinuities for generating physically smooth high-resolution air quality inference results.
- Extensive experiments show state-of-the-art accuracy on major air pollutants and remarkable zero-shot cross-region generalization, validating the proposed paradigm's superiority.

For clarity, a summary of the key notations used throughout this paper is provided in Table 1. The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the proposed model. Section 4 present the experimental setup and results. Section 5 concludes the paper.

2. Related work

2.1. Multi-source heterogeneous data fusion

Multi-source data fusion involves the integration of data from different sources, such as sensor data with different storage structures and data formats (Cheng et al., 2025; Wang et al., 2024b; Zheng et al., 2023b). The key challenge is the effective extraction and exploitation of the critical information that is embedded in these different types of data. Multi-source heterogeneity is similar to multimodal data, but has a broader scope, including structured, semi-structured, and unstructured data, and thereby covers a wider variety of data types (Zhang et al., 2018).

With the advent of the Big Data era, vast amounts of data are being generated in sectors ranging from social media to transportation, healthcare to wireless communication networks (John et al., 2021; Sun et al., 2022). This has led to research into using heterogeneous data from multiple sources to improve tasks such as urban planning, tackling air pollution and providing more accurate recommendations to users.

A deep learning-based method (Liu et al., 2020) was developed for urban data fusion, providing a systematic categorization and analysis of approaches to integrate multi-source urban data. Zhao et al. (2022) proposed a multi-source heterogeneous data fusion approach for ecological risk assessment and planning in the Songhua River Basin. Sun et al. (2024) designed a fire detection algorithm based on multi-source heterogeneous data. Wang et al. (2024c) utilized multi-source heterogeneous data fusion for tunnel traffic accident risk prediction. Qian et al. (2018) discuss its application in atmospheric environmental monitoring. Zhang et al.'s work (Zhang et al., 2022b) applied multi-source heterogeneous data for the interpolation and prediction of $PM_{2.5}$ concentrations. Yun and Luo (2022) designed a multi-source information fusion algorithm to achieve personalized travel destination recommendations.

2.2. Air quality inference

The air quality inference task is designed to provide air quality forecasts for the region of interest using a limited set of data from air stations. Spatial regression methods incorporate spatial correlation into statistical regression (Cressie, 2015; Miller, 2004). Spatial interpolation estimates the values at unknown locations based on the known sensor readings at specific positions. Inverse distance weighting (Wong et al., 2004) aggregates observation using a distance decay factor. In contrast, ordinary Kriging (Li & Heap, 2011) learns weights by minimizing the variance of estimation errors. The main issue with interpolation methods is that their performance tends to degrade significantly when air quality varies nonlinearly across geographic locations. Land-use-regression models (Jutzeler et al., 2014; Lin et al., 2017; Patel et al., 2022; Wang et al., 2019) infer air quality at unknown locations by learning the relationship between air quality and a set of land use variables (e.g., land use type, population density, and terrain elevation).

Sparse and low-rank methods aim to reconstruct high-dimensional data from limited measurements by exploiting the sparse or low-rank properties of the data. Wu et al. (2020) proposed an inference model based on Bayesian compressed sensing, which infers fine-grained pollution maps by leveraging calibrated air quality observations obtained from sensors deployed on shared bicycles. Shang et al. (2014) introduced a matrix completion method to estimate real-time pollution emissions on each road segment by utilizing traffic speed and flow learned from taxi trajectory data. Xu and Zhu (2016), Xu et al. (2019) proposed a context-aware tensor decomposition method for inferring fine-grained air pollution.

Semi-supervised learning methods use the underlying structure in unlabelled data to compensate for the lack of supervision. U-air (Zheng et al., 2013) infers urban air pollution information based on air quality data from measurement stations and heterogeneous urban data. Hsieh et al. (2015) used label propagation for urban air quality inference. Zhao et al. (2017) integrate a regularization strategy into a multi-task regression model for air quality inference.

In recent years, deep learning has been applied in air quality inference. AQNet (Rowley & Karakus, 2023) was proposed to integrate real-time remote sensing data, but often suffers from missing data. Ma et al. (2020) proposed a weather-aware deep autoencoder framework to infer fine-grained air pollution based on sparse observation data collected from mobile sensing systems. ADAIN (Cheng et al., 2018) is a neural attention-based model using observations from existing measurement stations to estimate air quality at any location in a given time period. MCAM (Han et al., 2021) is a multi-channel attention model to explicitly decouple the static and dynamic aspects of spatial correlation. STGNP (Hu et al., 2023a) introduces Graph Bayesian Aggregation, extending neural processes to spatio-temporal graph modeling. Patel et al. (2022) developed a Gaussian process-based method to estimate air quality. DualSTN (Hu et al., 2023b) proposes a dual spatial-temporal network to model long-range spatial and complex temporal patterns for accurate air quality inference from sparse station data. INCREASE (Zheng et al., 2023a) proposes an inductive graph repre-

sentation learning model for spatio-temporal kriging that jointly captures heterogeneous spatial relations and diverse temporal patterns. STFNN (Feng et al., 2024) introduces the concept of fields for air quality estimation. AirRadar (Wang et al., 2025) is a nationwide air quality inference model that integrates learnable mask tokens, and causal context adjustment to effectively address spatial correlation and heterogeneity.

3. Methodology

3.1. Overview

In order to make high-resolution spatio-temporal air quality inferences in sparsely monitored urban regions, it is necessary to reconstruct a physically continuous and temporally coherent pollution field from limited observations. Given a set of sparse monitoring stations $\mathcal{L}_{\text{obs}} = \{\mathbf{l}_1, \dots, \mathbf{l}_M\}$ with associated pollutant measurements $\{y_i\}$, and multi-source static data available across the entire domain $\mathcal{L}$, the objective is to infer the concentration $\hat{y}(\mathbf{l})$ at any unmonitored location $\mathbf{l} = (\text{lat}, \text{lon}) \in \mathcal{L} \setminus \mathcal{L}_{\text{obs}}$, such that the resulting field exhibits spatial smoothness, temporal consistency, and physical plausibility.

The architecture of SAQIN is illustrated in Fig. 1, which explicitly decouples the inference process into two parallel branches: a static branch that encodes long-term environmental priors from remote sensing and geospatial data, and a dynamic branch that models short-term spatio-chemical variations using multi-pollutant observations and geographic relationships. The SAQIN reformulates inference as a residual correction task by estimating the residual between the target location and reference stations, thereby avoiding the unknown global static bias and focusing on learning local dynamics. Finally, a stepwise multi-station weighted fusion mechanism is designed to eliminate Voronoi-induced discontinuities and generate a continuous and physically smooth air quality inference field with high spatial resolution.

3.2. Static-dynamic decoupling feature representation

Air quality inference should take the fundamental duality of its data sources into account, i.e., long-term static environmental priors and short-term dynamic pollution variation. Air quality at any location and time is jointly determined by persistent geographic and socioeconomic factors such as land use and population density, as well as short-term transient atmospheric processes including advection and chemical reactions. These two components do not act independently but interact in a nonlinear manner. The impact of dynamic processes is modulated by the underlying static context, and vice versa.

To characterize this duality in air quality inference, SAQIN explicitly decouples the representation of static and dynamic components, and thereafter enabling nonlinear fusion of these two components in the set-wise weighted fusion module. The observed pollutant concentration $y(\mathbf{l}, t)$ is modeled by

$$y(\mathbf{l}, t) = \mathcal{G}(\mathcal{F}_{\text{static}}(\mathbf{l}), \mathcal{F}_{\text{dynamic}}(\mathbf{l}, t)), \quad (1)$$

where $\mathcal{F}_{\text{static}}$ encodes the region's emission potential and dispersion capacity, $\mathcal{F}_{\text{dynamic}}$ captures the spatio-temporal evolution of pollution under meteorological and chemical influences, $\mathcal{G}$ is a learnable nonlinear fusion function. SAQIN has two parallel feature extraction branches. One branch characterizes the static contextual priors, and the other one captures dynamic spatio-temporal evolution. The static and dynamic features are subsequently fused in a residual correction module that implicitly learns the interaction between the static background and dynamic variations.

3.2.1. Target static feature representation

The static background field of air quality is governed by geographically stable factors that shape a region's long-term emission potential and atmospheric dispersion capacity. These factors including land use

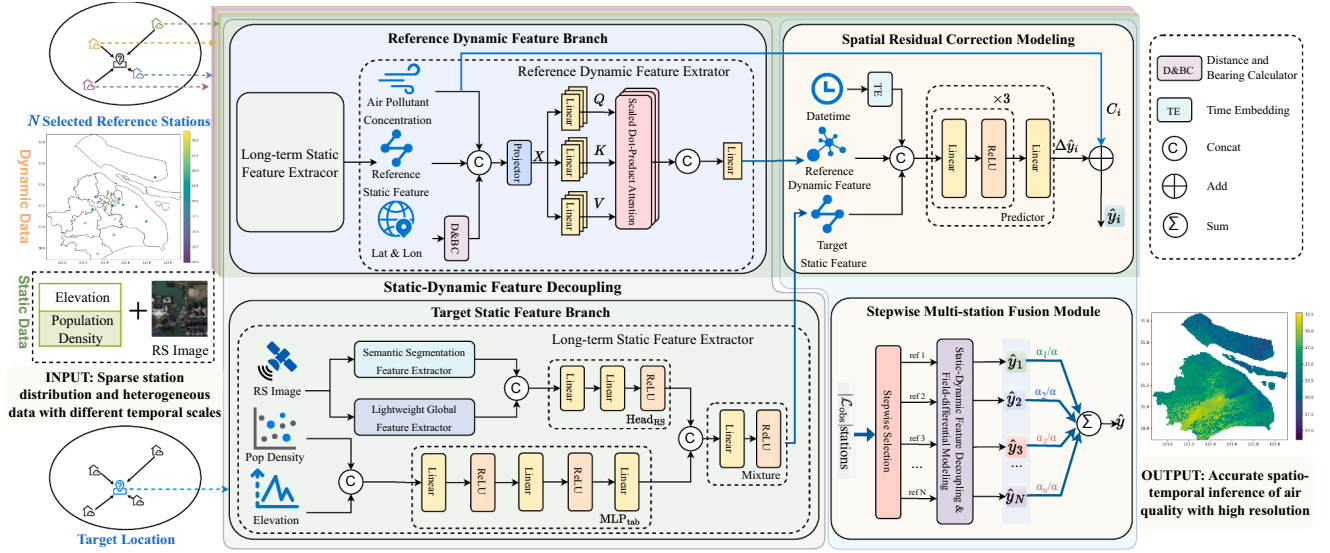


Fig. 1. The framework of SAQIN, which comprises a target static feature branch, a reference dynamic feature branch, a spatial residual correction module, and a stepwise multi-station fusion module.

patterns, population distribution, and topographic elevation are invariant at the temporal scale of air quality monitoring (hourly to daily) yet exhibit rich spatial heterogeneity that directly modulates pollutant concentrations. To represent this static background field, we propose a holistic static feature extraction framework that moves beyond pixel-level statistics or handcrafted indices toward a semantic-level, mechanism-aware representation of environmental priors.

The inherent logic of this framework is that remote sensing imagery encodes not just visual textures but interpretable surface semantics that are causally linked to pollution sources. For instance, industrial zones emit primary particulates and NO_x , dense road networks generate traffic-related CO and $\text{PM}_{2.5}$ (Karagülian et al., 2015), and green spaces act as sinks for certain pollutants. Prior approaches, however, typically consider satellite imagery as generic pixel inputs to standard convolutional networks or a coarse module like NDVI. Although standard CNNs can learn high-level representations, they are typically trained using end-to-end regression. This results in abstract latent representations that lack physical interpretability, which is shown in Fig. 2(b). Similarly, the NDVI (Fig. 2(c)) only captures vegetation, ignoring critical emission sources such as industrial belts.

In contrast, we leverage high-resolution semantic segmentation to explicitly recover pollution-relevant land-use categories from remote sensing imagery. Our module employs transfer learning based on explicit semantic supervision to provide physically meaningful environmental priors (Fig. 2(d)), which is different from generic texture-based feature extractors. Given the input remote sensing image I_{RS} , we employ DeepLabV3 (Chen et al., 2017) with a ResNet50 (He et al., 2016) backbone to extract high-level semantic features, such as built-up area, road, water, forest, and agriculture. We initialize the module using weights pre-trained on the LoveDA dataset (Wang et al., 2021), which provides pixel-level supervision for remote sensing semantic segmentation, enabling our model to leverage robust environmental priors without requiring manual annotation of the target region. The ResNet50 backbone processes I_{RS} through multiple stages, producing a hierarchy of feature maps $\mathcal{F}_{\text{res}}^{(1)}, \mathcal{F}_{\text{res}}^{(2)}, \dots, \mathcal{F}_{\text{res}}^{(N)}$ that capture spatial patterns at progressively coarser resolutions. These multi-level features are fed into the Atrous Spatial Pyramid Pooling (ASPP) module, which applies parallel atrous convolutions with different dilation rates to encode contextual information across multiple receptive fields. This enables the model to distinguish fine-grained land-use configurations, such as a small factory embedded in a rural area and a large industrial park, by jointly modeling local details and broader surroundings. The outputs of the ASPP

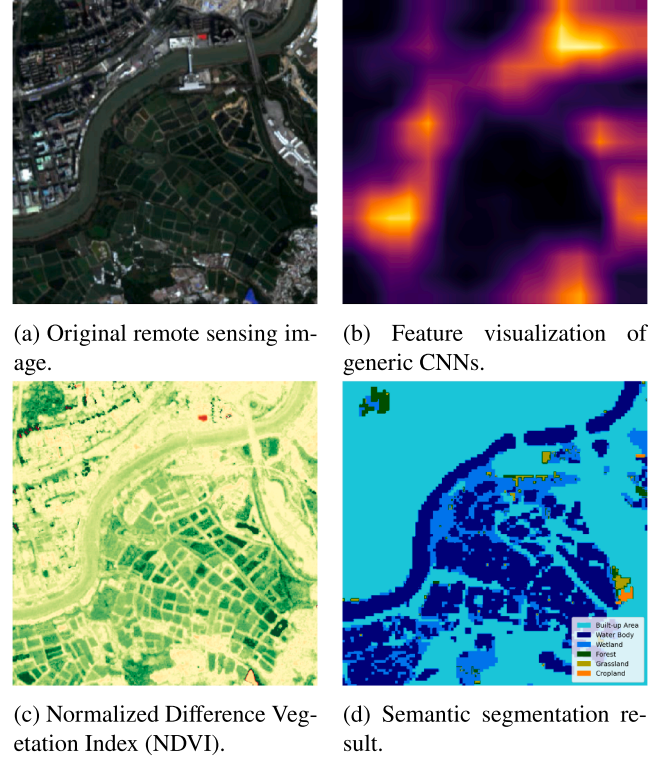


Fig. 2. Visual comparison of different remote sensing representation methods. (b) Generic CNN features typically yield abstract latent representations without clear physical meaning; (c) NDVI is limited in capturing vegetation information; (d) Our approach based on pre-trained semantic segmentation explicitly recovers fine-grained land-use categories (e.g., built-up area, water, forest, and agriculture), providing physically meaningful environmental prior.

branches are concatenated to form a fused feature map:

$$\mathcal{F}_{\text{aspp}} = \text{Concat}(\mathcal{F}_{\text{res}}^{(1)}, \mathcal{F}_{\text{res}}^{(2)}, \dots, \mathcal{F}_{\text{res}}^{(N)}). \quad (2)$$

This representation is then passed through a DeepLab head consisting of two fully connected layers, followed by global average pooling

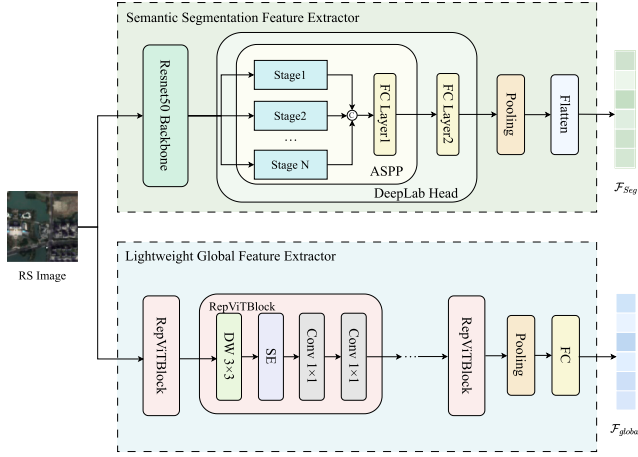


Fig. 3. Semantic segmentation and global features extraction from remote sensing images using a semantic segmentation feature extractor based on DeepLabV3 (Chen et al., 2017) and a lightweight global feature extractor based on RepViTBlock (Wang et al., 2024a).

and flattening to yield a compact semantic feature:

$$F_{\text{Seg}} = \text{Flatten}(\text{GAP}(\text{FC}_2(\text{FC}_1(F_{\text{aspp}})))) \quad (3)$$

Here, FC_1 and FC_2 denote two fully connected layers that project the ASPP features into an intermediate embedding space, GAP represents global average pooling which reduces each spatial channel to its mean value, and Flatten reshapes the resulting one-dimensional vector into a flat feature representation suitable for downstream fusion. This design leverages high-resolution semantic segmentation to explicitly model land-use categories that are closely associated with pollution sources, thereby integrating fine-grained visual understanding from computer vision into environmental air quality modeling. The detailed architecture of the Semantic Segmentation Feature Extractor is depicted in the upper half of Fig. 3.

In addition to local semantics, the static background field is also shaped by large-scale spatial structures, such as urban cores, suburban belts, and rural hinterlands, that govern regional airflow, heat island effects, and emission density gradients. To capture these macro-scale patterns without incurring the computational cost of vision transformers, we introduce a lightweight global context extractor based on M stacked RepViTBlocks (Wang et al., 2024a).

Starting from the input remote sensing image I_{RS} , the extractor processes the image through M sequential blocks. Each block performs three operations in sequence: (i) a 3×3 depthwise convolution for spatial mixing; (ii) a Squeeze-and-Excitation (SE) layer for channel-wise feature recalibration; (iii) two 1×1 convolutions for representation refinement. The transformation within the m_{th} block is defined by

$$\mathcal{X}^{(m)} = \text{Conv}_{1 \times 1}^{(2)} \left(\sigma(\text{MLP}(\text{GAP}(\text{Conv}_{\text{DW}}(\mathcal{X}^{(m-1)})))) \right) \odot \text{Conv}_{\text{DW}}(\mathcal{X}^{(m-1)}) \quad (4)$$

where $\mathcal{X}^{(m)}$ denotes the m_{th} feature map after the stem layer, Conv_{DW} represents the depthwise convolution, MLP is a two-layer perceptron, σ is the sigmoid function, and $\odot$ denotes element-wise multiplication. The output after M blocks, denoted $\mathcal{X}^{(M)}$, possesses a global receptive field while maintaining low parameter count. This feature map is then processed by global average pooling followed by a fully connected layer to yield a compact global contextual feature:

$$F_{\text{global}} = \text{FC}(\text{GAP}(\mathcal{X}^{(M)})) \quad (5)$$

The detailed architecture of the Lightweight Global Feature Extractor is illustrated in the lower half of Fig. 3.

This design efficiently encodes macro-scale spatial configurations that complement the fine-grained semantics from the segmentation

branch, enabling SAQIN to jointly model both local pollution sources and regional environmental contexts. The two branches are fused to form a comprehensive remote sensing representation:

$$F_{\text{RS}} = \text{Head}_{\text{RS}}(\text{Concat}(F_{\text{Seg}}, F_{\text{global}})), \quad (6)$$

where Head_{RS} is a lightweight projection network consisting of two linear layers with ReLU activation.

Finally, we encode the structured geospatial variables, population density ρ and elevation h , into a tabular feature vector using a multilayer perceptron:

$$F_{\text{tab}} = \text{MLP}_{\text{tab}}((\rho, h)), \quad (7)$$

where MLP_{tab} consists of three linear layers with ReLU activations in between. Population density reflects the intensity of anthropogenic activities such as traffic volume and industrial emissions. Elevation affects vertical atmospheric mixing and can lead to pollutant trapping in topographically constrained regions.

The final static feature is obtained by fusing the remote sensing representation F_{RS} and the tabular feature F_{tab} through a lightweight fusion block composed of linear layers and ReLU activations:

$$F_{\text{static}} = \text{Mixture}(\text{Concat}(F_{\text{RS}}, F_{\text{tab}})). \quad (8)$$

This integrated representation captures complementary aspects of the static background field, enabling the model to distinguish between regions that exhibit similar pollutant concentrations but originate from different emission sources, such as a high- $\text{PM}_{2.5}$ industrial zone and a high- $\text{PM}_{2.5}$ traffic corridor. Such distinctions are difficult for models that conflate static and dynamic features without explicit decoupling.

3.2.2. Reference dynamic feature representation

While the static background field captures long-term environmental priors, the dynamic perturbation field encodes the short-term, non-stationary processes that drive air quality fluctuations. These perturbations arise from two intertwined physical mechanisms: (i) atmospheric transport, where pollutants are advected and dispersed by wind and turbulence; and (ii) chemical transformation, where primary pollutants undergo photochemical reactions to form secondary species. To represent this dynamic evolution, we design a feature aggregation mechanism that jointly models geographic transport pathways and inter-pollutant chemical couplings.

For each reference monitoring station, we construct a comprehensive feature vector that captures the essential dimensions of atmospheric perturbation dynamics through complementary sources of information.

The core of this representation lies in real-time concentrations of major pollutants: CO, NO₂, O₃, PM₁₀ and PM_{2.5}. This multi-pollutant input is not merely a data augmentation strategy but a direct reflection of atmospheric chemistry. For instance, NO₂ and O₃ participate in a well-known photochemical cycle (Seinfeld & Pandis, 2016): NO₂ photolyzes under sunlight to produce O₃, while O₃ reacts with NO to regenerate NO₂. Similarly, PM_{2.5} and PM₁₀ share common sources such as combustion and dust, leading to strong co-emission patterns. As shown in Fig. 4, the empirical correlation structure reveals a strong positive coupling between PM_{2.5} and PM₁₀ ($r = 0.74$) and a moderate negative coupling between NO₂ and O₃ ($r = -0.48$). By jointly modeling all pollutants, our framework leverages these inherent chemical interdependencies to infer the state of the entire atmospheric system, rather than treating each species in isolation.

Complementing chemical signals, we encode the geometric relationship between the reference station and the target location using precise geodesic computations. The Haversine formula computes the great-circle distance d :

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos\phi_1 \cos\phi_2 \sin^2\left(\frac{\Delta\lambda}{2}\right), \quad (9)$$

$$c = 2 \cdot \arctan 2(\sqrt{a}, \sqrt{1-a}), \quad (10)$$

$$d = R \cdot c, \quad (11)$$

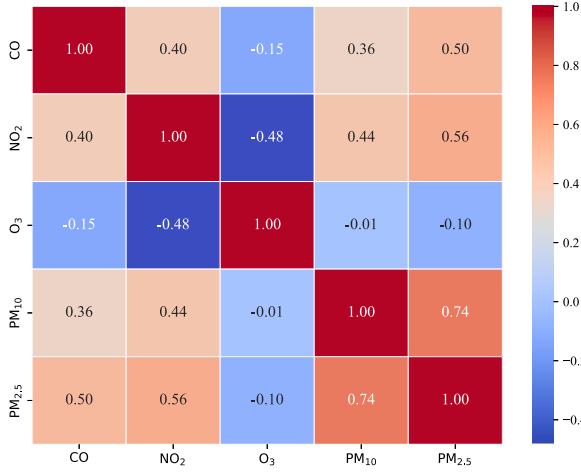


Fig. 4. Correlation matrix of five air pollutants at Hongkou Station (121.467°E, 31.3008°N; Shanghai, China), January 1, 2021 - November 16, 2024. Colours represent the Pearson correlation coefficients (r), where red ($r \approx +1$), blue ($r \approx -1$), and white ($r \approx 0$) indicate positive, negative, and no linear correlations, respectively.

where ϕ_1, ϕ_2 are latitudes, λ_1, λ_2 are longitudes, $\Delta\phi = \phi_2 - \phi_1$, $\Delta\lambda = \lambda_2 - \lambda_1$, and $R = 6371$ km is Earth's mean radius. Crucially, we also compute the bearing θ :

$$v = \sin(\Delta\lambda) \cos \phi_2, \quad (12)$$

$$u = \cos \phi_1 \sin \phi_2 - \sin \phi_1 \cos \phi_2 \cos(\Delta\lambda), \quad (13)$$

$$\theta = \arctan 2(v, u). \quad (14)$$

The bearing θ encodes the directional relationship between the reference station and the target location. This geometric cue allows the model to learn anisotropic spatial dependencies, such as distinguishing stations that lie upwind versus downwind under prevailing wind patterns, even when explicit meteorological data is unavailable.

Finally, to ensure contextual consistency between the static background and dynamic perturbation fields, we incorporate the reference station's own static feature $\mathcal{F}_{\text{static}}^{(i)}$. This allows the model to interpret dynamic observations relative to their local environmental setting, for example by recognizing that a given NO_2 concentration in an industrial zone implies a different emission regime than the same value in a residential neighborhood.

For each reference station, the feature vector $\mathbf{x}_i$ is constructed as

$$\mathbf{x}_i = \text{Concat} \left(C_{\text{CO}}, C_{\text{NO}_2}, C_{\text{O}_3}, C_{\text{PM}_{10}}, C_{\text{PM}_{2.5}}, d_i, \theta_i, \mathcal{F}_{\text{static}}^{(i)} \right), \quad (15)$$

where C denotes real-time pollutant concentrations, d_i and θ_i are the distance and bearing from target location to the i_{th} reference station, and $\mathcal{F}_{\text{static}}^{(i)}$ is the static feature of the i_{th} reference station. This vector is then projected into a sequence of latent tokens $\mathbf{X} \in \mathbb{R}^{L \times d}$, which is processed by a multi-head self-attention mechanism. The query Q , key K , and value V are obtained through shared linear projections:

$$Q = \mathbf{X}W^Q, \quad K = \mathbf{X}W^K, \quad V = \mathbf{X}W^V, \quad (16)$$

where $W^Q, W^K, W^V \in \mathbb{R}^{d \times d_k}$ are learnable weight matrices and d_k is the head dimension. These are split into h heads, and the attention output for the j_{th} head is computed as

$$h_j = \text{softmax} \left(\frac{Q_j K_j^T}{\sqrt{d_k}} \right) V_j, \quad (17)$$

where $Q_j, K_j, V_j \in \mathbb{R}^{L \times d_k}$ denote the j_{th} head of Q, K, V , respectively. The outputs of all heads are concatenated and projected through a final linear layer to produce the dynamic feature representation:

$$\mathcal{F}_{\text{dynamic}}^{(i)} = \text{Concat}(h_1, \dots, h_h)W^O, \quad (18)$$

where $W^O \in \mathbb{R}^{d \times d_{\text{out}}}$ is a learnable output projection matrix.

This self-attention mechanism dynamically models the contextual influence of each reference station through a learned compatibility function that jointly incorporates geographic proximity, relative direction, and inter-pollutant coupling patterns. This formulation captures the non-stationary and anisotropic nature of real-world pollution dispersion, where a station's relevance depends not only on its distance to the target but also on its directional alignment and the spatio-chemical relationships among pollutants.

By explicitly modeling the dynamic perturbation field as a spatio-chemical process, SAQIN moves beyond naive spatial interpolation to a physically grounded representation of atmospheric dynamics.

3.3. Spatial residual correction modeling

Conventional air quality inference models formulate the task as a standard regression problem: given auxiliary features at an unmonitored location $\mathbf{l}$, predict its absolute pollutant concentration $y(\mathbf{l}, t)$. Note that the observed concentration at any location can be modeled as the superposition of two distinct components, i.e.,

$$y(\mathbf{l}, t) = f_{\text{bg}}(\mathbf{l}) + \delta f(\mathbf{l}, t), \quad (19)$$

where $f_{\text{bg}}(\mathbf{l})$ is a spatially coherent background field shaped by persistent environmental factors such as land use and population density, and $\delta f(\mathbf{l}, t)$ is a locally varying perturbation field driven by transient atmospheric processes including wind advection and photochemical reactions. The background field $f_{\text{bg}}(\mathbf{l})$ is inherently non-stationary across large spatial scales, as it reflects the heterogeneous distribution of emission sources and land use patterns (Hoek et al., 2008; Jerrett et al., 2005). In contrast, the perturbation field $\delta f(\mathbf{l}, t)$ characterizes atmospheric diffusion processes governed by fluid dynamics (Seinfeld & Pandis, 2016), which enforce physical continuity and thus exhibit local smoothness (Pasquill & Smith, 1983; Tobler, 1970).

Direct regression on $y(\mathbf{l}, t)$ conflates these two physically distinct components, leading to models that capture spurious correlations and fail to generalize across regions with differing baselines. To resolve this entanglement, SAQIN reformulates air quality inference as a residual correction task grounded in multi-scale spatial decomposition. In a sparse-monitoring regime, the large-scale background field $f_{\text{bg}}(\mathbf{l})$ is poorly constrained due to limited observations, whereas the small-scale perturbation field $\delta f(\mathbf{l}, t)$ exhibits strong local smoothness. Consequently, the relative perturbation between nearby locations tends to be small and predictable.

Leveraging this insight, SAQIN estimates the concentration offset relative to the i_{th} reference station with observed concentration $C_i = y(\mathbf{l}_i, t)$:

$$\Delta y_i(\mathbf{l}, t) = y(\mathbf{l}, t) - C_i. \quad (20)$$

Substituting the field decomposition yields

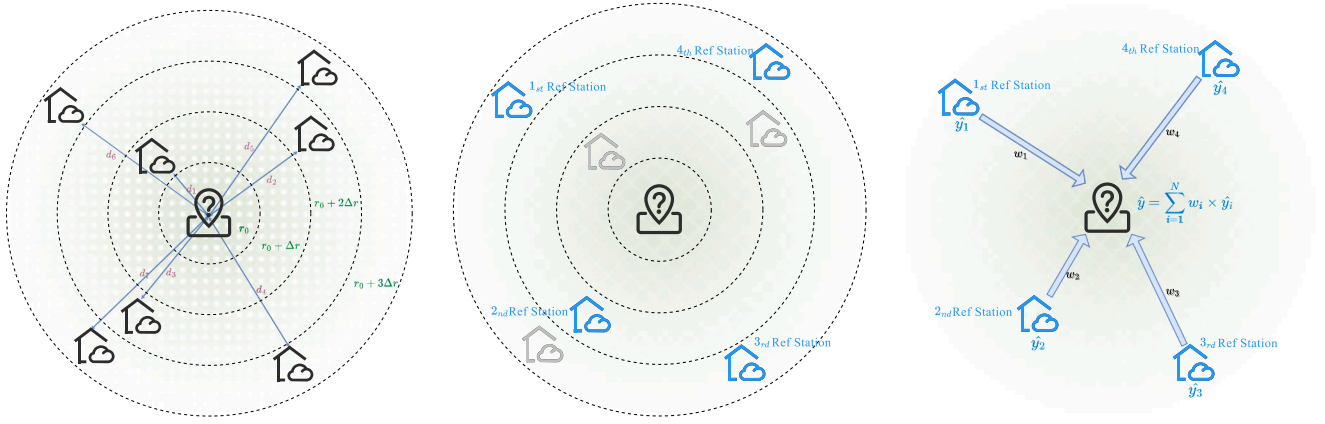
$$\Delta y_i(\mathbf{l}, t) = [f_{\text{bg}}(\mathbf{l}) + \delta f(\mathbf{l}, t)] - [f_{\text{bg}}(\mathbf{l}_i) + \delta f(\mathbf{l}_i, t)]. \quad (21)$$

Under the assumption of local stationarity for geographically proximate locations, the background field varies slowly, so that $f_{\text{bg}}(\mathbf{l}) \approx f_{\text{bg}}(\mathbf{l}_i)$. The offset then simplifies to

$$\Delta y_i(\mathbf{l}, t) = \delta f(\mathbf{l}, t) - \delta f(\mathbf{l}_i, t). \quad (22)$$

This formulation implicitly normalizes out the unknown large-scale background by conditioning on physically measured values, allowing the model to focus learning exclusively on the locally smooth perturbation dynamics. The term residual here refers not to model prediction error but to this physically grounded concentration offset that captures local atmospheric dynamics.

To predict this deviation, the offset predictor Φ fuses three physically grounded signals. The static environmental context of the target location $\mathcal{F}_{\text{static}}$ encodes the persistent background that modulates local pollution potential. The spatio-chemical state of the reference station $\mathcal{F}_{\text{dynamic}}^{(i)}$



(a) Select $|\mathcal{L}_r|$ air monitoring stations using Stepwise Selection Strategy. (b) Randomly select N stations from $\mathcal{L}_r$ as reference stations during training phase. (c) Fuse the offset predictions from the reference stations.

Fig. 5. Illustration of the stepwise multi-station fusion framework: (a) stepwise selection of neighboring monitoring stations, (b) random selection of N reference stations from $\mathcal{L}_r$ during the step of training, and (c) fusion of offset predictions $\hat{y}_i$ obtained from the reference stations.

captures transient perturbation dynamics, including multi-pollutant interactions and geographic influences. A deterministic time embedding $\mathbf{P}_T$ encodes multi-scale temporal periodicities without trainable parameters. Discrete time step T is represented as a tuple (year, month, day, weekday, hour), with each component mapped to a continuous vector through sine and cosine functions of predefined periods. The m_{th} dimension of $\mathbf{P}_T$ is given by

$$\mathbf{P}_{(T,m)} = \begin{cases} \sin(2\pi t_k/T_k) & \text{if } m \bmod 2 = 0, \\ \cos(2\pi t_k/T_k) & \text{if } m \bmod 2 \neq 0, \end{cases} \quad (23)$$

where $k = \lfloor m/2 \rfloor + 1$, t_k is the k_{th} time component, and the period set $T = \{10, 12, 31, 7, 24\}$ corresponds to annual, monthly, daily, weekly, and hourly cycles, respectively.

The concentration offset relative to the i_{th} reference station is predicted as

$$\Delta \hat{y}_i(\mathbf{l}, t) = \Phi(\text{Concat}(\mathcal{F}_{\text{static}}, \mathcal{F}_{\text{dynamic}}^{(i)}, \mathbf{P}_T)), \quad (24)$$

and the final inferred concentration is reconstructed via a residual update:

$$\hat{y}_i(\mathbf{l}, t) = C_i + \Delta \hat{y}_i(\mathbf{l}, t). \quad (25)$$

This formulation anchors predictions to physically measured values and mitigates the extrapolation bias inherent in direct concentration regression. By decoupling large-scale background and small-scale perturbation at the inference level, SAQIN achieves much better statistical stability and generalization capability.

3.4. Continuous reconstruction with stepwise multi-station fusion

Existing air quality inference methods typically treat monitoring stations as discrete anchors and perform interpolation or regression based on these fixed points. However, atmospheric pollution is inherently a continuous physical field, with concentrations varying smoothly across space due to diffusion and transport processes. Artificial discontinuities should not arise simply because monitoring stations are sparsely distributed. The discrete-anchor modeling paradigm, by associating each unobserved location with one or a fixed set of reference stations, implicitly partitions the domain into Voronoi-like regions, leading to abrupt prediction jumps at cell boundaries—violating the physical smoothness of real atmospheric fields.

Inspired by the concept of implicit neural representations, particularly the Local Implicit Image Function (LIIF) framework (Chen et al.,

2021) which models an image as a continuous function, we reformulate air quality field inference as a continuous implicit field learning problem. The core idea is that each monitoring station provides not merely a scalar concentration value, but defines a local implicit function that characterizes the pollution field in its vicinity. The concentration at any target location is then obtained by fusing the outputs of multiple such local implicit functions from its neighboring stations.

To realize this principle, we propose a stepwise multi-station fusion mechanism. For any target location, a set of reference stations is dynamically selected via stepwise radius expansion: starting from an initial radius of 25 km centered at the target, the search radius is incrementally increased by 25 km until at least N candidate stations are found. During training, N stations are randomly sampled from this candidate set to promote diversity in reference configurations and enhance model robustness. During inference, the N nearest stations are deterministically selected to ensure prediction stability. This adaptive selection strategy guarantees that each target receives sufficient and representative local field information while balancing stochastic exploration and deterministic exploitation. The complete procedure is detailed in Algorithm 1.

Given these N reference stations, we fuse their individual offset predictions through a distance-weighted averaging scheme. This design is guided by the spatial statistics of atmospheric pollution: empirical studies under neutral atmospheric stability and urban conditions show that the spatial correlation of pollutant concentrations decays approximately as a power law $C \propto 1/d^k$ with $k \approx 0.5$ (Arya et al., 1999; Pasquill, 1996). Accordingly, the weight for the i_{th} station is defined as $\alpha_i = 1/\sqrt{d_i}$, where d_i is the geodesic distance to the target location. This weighting scheme ensures that nearby stations, whose local implicit fields are more reliable for capturing fine-scale perturbations, dominate the fused prediction, while distant stations contribute contextual smoothing. The resulting aggregation produces a physically smooth and continuous air quality field. The final inferred concentration at the target location $\mathbf{l}$ is given by

$$\hat{y}(\mathbf{l}, t) = \sum_{i=1}^N \frac{\alpha_i}{\alpha} \cdot \hat{y}_i(\mathbf{l}, t), \quad (26)$$

where $\alpha = \sum_{i=1}^N \alpha_i$. The entire Stepwise Multi-Station Fusion process is illustrated in Fig. 5.

Analogous to LIIF that treats pixels as samples of a continuous function over 2D coordinates, SAQIN treats monitoring stations as observational samples of a continuous pollution field over geographic space. By fusing multiple local implicit fields, i.e., the offset predictions from

Algorithm 1: Stepwise selection of reference stations.

Input : Target location $\mathbf{l} = (\text{lat}_t, \text{lon}_t)$;
 Station set $\mathcal{L}_{obs} = \{\mathbf{l}_i\}_{i=1}^M$;
 Initial radius $r_0 = 25$ km;
 Radius increment $\Delta r = 25$ km;
 Desired count $N = 4$;
 Training flag $\text{is_training} \in \{\text{True}, \text{False}\}$
Output: Reference station set $\mathcal{L}_{ref}^{(N)} \subseteq \mathcal{L}_{obs}$

```

1  $r \leftarrow r_0$ ;
2 while True do
3   Compute distances  $d_i = \text{dist}(\mathbf{l}, \mathbf{l}_i)$  for all  $i = 1, \dots, M$ ;
4    $\mathcal{L}_r \leftarrow \{\mathbf{l}_i \in \mathcal{L}_{obs} \mid d_i \leq r\}$ ;
5   if  $|\mathcal{L}_r| \geq N$  then
6     break;
7   end
8    $r \leftarrow r + \Delta r$ ;
9 end
10 if is_training then
11   Randomly select  $N$  stations from  $\mathcal{L}_r$ ;
12    $\mathcal{L}_{ref}^{(N)} \leftarrow$  selected stations;
13 end
14 else
15   Sort  $\mathcal{L}_r$  by  $d_i$  in ascending order;
16    $\mathcal{L}_{ref}^{(N)} \leftarrow$  first  $N$  stations in sorted list;
17 end
18 return  $\mathcal{L}_{ref}^{(N)}$ ;
```

individual reference stations, SAQIN reconstructs a globally continuous and physically smooth air quality field at any unmonitored location. This mechanism effectively alleviates the spatial discontinuities in traditional discrete modeling approaches, offering a practical pathway for high-resolution environmental field inference.

3.5. Loss function

To address the multi-scale, long-tailed, and heterogeneous nature of air quality data, we propose a composite loss function that jointly optimizes three complementary objectives: Smooth L1 Loss $\mathcal{L}_{\text{smooth}}$ (Ren et al., 2015), Log-L1 Loss $\mathcal{L}_{\text{log}}$, and Scale-Invariant Loss $\mathcal{L}_{\text{scale}}$. Each component targets a distinct aspect of the prediction error distribution, thereby improving model robustness, accuracy, and generalization across diverse environmental conditions.

The Smooth L1 Loss combines the benefits of L1 and L2 losses by being quadratic for small residuals and linear for large residuals. This ensures stable gradients near the optimum while limiting the influence of outliers, which is critical for modeling rare but severe pollution events such as winter haze episodes. It is defined as

$$\mathcal{L}_{\text{smooth}}(y, \hat{y}) = \begin{cases} 0.5 \cdot (\hat{y} - y)^2 & \text{if } |\hat{y} - y| < 1, \\ |\hat{y} - y| - 0.5 & \text{otherwise.} \end{cases} \quad (27)$$

The Log-L1 Loss applies a logarithmic transformation to the absolute prediction error. This reduces the penalty for large deviations while emphasizing relative accuracy in low-to-moderate concentration ranges, which correspond to typical background or mild pollution levels. By compressing the error scale, this loss encourages the model to prioritize performance in the most frequently observed regimes, leading to more stable convergence. Defined by

$$\mathcal{L}_{\text{log}}(y, \hat{y}) = \frac{1}{N} \sum_{i=1}^N \log(1 + |\hat{y}_i - y_i|). \quad (28)$$

The Scale-Invariant Loss measures prediction error in the logarithmic domain to capture relative discrepancies between predicted and



Fig. 6. Sentinel-2 satellite image of the Yangtze River Delta region with red markers indicating the locations of air monitoring stations.

ground-truth values. Although air quality concentrations are not inherently scale-invariant, they exhibit substantial spatial and temporal variability, often spanning multiple orders of magnitude across industrial and residential areas or between seasons. By penalizing the squared difference of log-transformed values, this loss promotes consistent relative prediction accuracy and enhances generalization across heterogeneous monitoring contexts:

$$\mathcal{L}_{\text{scale}}(y, \hat{y}) = \frac{1}{N} \sum_{i=1}^N (\log(\hat{y}_i + \epsilon) - \log(y_i + \epsilon))^2, \quad (29)$$

where $\epsilon = 10^{-6}$ is a small constant that ensures numerical stability for near-zero concentrations. The overall loss function is defined as the weighted sum of three components:

$$\mathcal{L}(y, \hat{y}) = \lambda_1 \mathcal{L}_{\text{smooth}}(y, \hat{y}) + \lambda_2 \mathcal{L}_{\text{log}}(y, \hat{y}) + \lambda_3 \mathcal{L}_{\text{scale}}(y, \hat{y}), \quad (30)$$

with $\lambda_1, \lambda_2, \lambda_3 > 0$. Preliminary sensitivity analysis shows that the three loss terms operate on comparable numerical scales and maintain stable gradient magnitudes throughout training. We adopt weights $\lambda_1 = \lambda_2 = \lambda_3 = 1$, which provides a balanced trade-off among robustness to extreme events, accuracy under normal conditions, and consistency across magnitude scales.

4. Experiments

4.1. Implementation details

Our experiments are conducted on a single NVIDIA RTX 3090, and the network framework is built using PyTorch. During training, we employ the AdamW (Loshchilov, 2017) optimizer with an initial learning rate of 5×10^{-4} , and the momentum coefficients β_1 and β_2 are set to 0.9 and 0.95, respectively. To further improve the training performance, we introduce a Cosine Annealing Learning Rate Scheduler (Loshchilov & Hutter, 2016). The batch-size during training is set to 32. We set the number of reference stations N to be 4.

4.2. Datasets description

We conduct experiments on the dataset of the Yangtze River Delta region, which comprises 171 monitoring stations. The information of the dataset are shown in Fig. 6 and Table 2. Open-source data is utilized for model training. The air quality data for the Yangtze River Delta region comes from the China National Environmental Monitoring Center,¹ remote sensing data is obtained from the European Space Agency's Sentinel-2 satellite, which provides high-resolution multispectral images, accessible via Google Earth Engine.² Population density data is

¹ <https://air.cnemc.cn:18007/>

² <https://code.earthengine.google.com/>

Table 2
Dataset information for the Yangtze river delta.

Description	Value
Longitude Range	116.80°E - 122.60°E
Latitude Range	29.60°N - 32.80°N
Number of Stations	171
Training Period	1/1/2021 - 12/31/2023
Testing Period	1/1/2024 - 11/16/2024

Table 3
Resolutions of different data sources.

Data Type	Resolution
Sentinel-2 Satellite Imagery	10m × 10m
Population Density	1km × 1km
Elevation	0.5km × 0.5km

retrieved from WorldPop,³ and elevation data comes from GEBCO.⁴ The resolutions of the visible satellite image, population density, and elevation data are summarized in Table 3. Since monitoring station data is occasionally lost, we use linear interpolation to fill the missing data. All features are normalized to ensure consistent scales across different data sources. In addition to the primary dataset, we validate the proposed model on two more datasets, i.e., using Beijing dataset with 32 stations to verify cross-domain generalization and Shenzhen dataset with 15 stations for ablation studies of the proposed modules and loss function.

4.3. Evaluation metrics

We use four metrics to assess the performance of our model, i.e., Coefficient of determination (R^2), Mean Absolute Error (MAE), Mean Relative Error (MRE), and Root Mean Squared Error (RMSE). R^2 measures the closeness between the predicted values and the ground-truth values. MAE represents the average absolute difference between the predicted and ground-truth values. MRE calculates the average ratio of the prediction error to the ground-truth value, reflecting the relative magnitude of the prediction error. RMSE calculates the square root of the average squared difference between the predicted and ground-truth values. These four metrics are defined as follows.

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}, \quad (31)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|, \quad (32)$$

$$MRE = \frac{1}{N} \sum_{i=1}^N \frac{|y_i - \hat{y}_i|}{|y_i|}, \quad (33)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}, \quad (34)$$

where $\bar{y}$ is the mean of the ground-truth values.

4.4. Baseline methods

We evaluate SAQIN against the following baselines, grouped into three categories:

- **Nonparametric methods:** KNN (Baran, 2021; Guo et al., 2003; Marchang & Tripathi, 2020), which predicts via nearest-neighbor aggregation. GPR (Williams & Rasmussen, 1995), a Gaussian process regression model capturing spatio-temporal correlations through kernels.

- **Kriging-based methods:** IGNNK (Wu et al., 2021), an inductive graph neural network for kriging that generalizes to unsampled locations via subgraph-based message passing with dynamic adjacency reconstruction. INCREASE (Zheng et al., 2023a), an inductive graph model leveraging multi-relation learning for spatio-temporal kriging.
- **Deep spatio-temporal methods:** STGNP (Hu et al., 2023a), a Bayesian graph-based method modeling uncertainty and relational structure. DualSTN (Hu et al., 2023b), which decouples short- and long-term temporal dependencies via a joint spatiotemporal attention network and a graph recurrent module with time-skip connections. STFNN (Feng et al., 2024), which leverages the Euclidean continuity of air quality for spatio-temporal forecasting. Air-Radar (Wang et al., 2025), a framework using masked reconstruction and causal context to improve spatial generalization.

4.5. Inference visualization

Fig. 7 shows the inference result of air pollutant concentrations using the SAQIN model in Shanghai, which is part of the Yangtze River Delta. The concentration distributions of NO_2 , $PM_{2.5}$, PM_{10} , and CO exhibit a significant positive correlation, while demonstrating an inverse relationship with O_3 concentrations, which validates the observation drawn from Fig. 4.

4.6. Model comparison

We evaluate all models on the Yangtze River Delta dataset, with results in Table 4. Classical methods KNN and GPR show limited accuracy, especially for CO, where R^2 is below 0.32, reflecting their inability to capture nonlinear pollution dynamics. Graph-based kriging methods IGNNK and INCREASE improve spatial generalization through inductive graph learning. INCREASE achieves the highest performance among these, yet still exhibits reduced accuracy on CO, indicating persistent challenges in modeling emission heterogeneity.

Among deep spatio-temporal models, STGNP balances spatial and temporal dependencies, while STFNN leverages field continuity. Both treat concentration as a direct regression target, leading to conflation of static and dynamic components. AirRadar improves generalization through masked reconstruction, achieving strong results on O_3 , NO_2 , $PM_{2.5}$, and PM_{10} , but remains limited on CO. DualSTN decouples temporal scales and shows improved capability on CO, yet its spatial modeling is constrained by discrete station anchors.

Our proposed SAQIN framework consistently outperforms all baselines across all five pollutants. It achieves R^2 scores of 0.9474 for O_3 , 0.8842 for NO_2 , 0.9542 for $PM_{2.5}$, 0.9509 for PM_{10} , and 0.6485 for CO. SAQIN reduces MAE by up to 25% and RMSE by up to 30% compared to AirRadar and STGNP. This improvement stems from its mechanism-aligned design: by modeling residual offsets relative to reference station observations, it explicitly disentangles the unknown static baseline from the locally smooth dynamic perturbation. Stepwise multi-station fusion with distance-weighted averaging then reconstructs a continuous, physically plausible air quality field, eliminating artificial discontinuities inherent in discrete-anchor methods.

We conduct comparisons with of deep spatio-temporal learning methods on the $PM_{2.5}$ dataset, which is shown in Fig. 8. Throughout the evaluation period, SAQIN consistently achieves the highest R^2 scores. This indicates a stronger correlation between its predictions and ground truth observations from air quality monitoring stations. Baseline models such as STGNP and STFNN show fluctuating or negative R^2 values on certain days, reflecting instability or a lack of generalization. Regarding the MAE, SAQIN shows the lowest errors throughout the year, especially during the high pollution periods in winter and early spring. This suggests that our model is more robust to variations in the level of air pollution. In addition, the MRE curves show that SAQIN also maintains a smaller error relative to the magnitude of the ground truth concentrations, underlining its stability over different pollution scales. Finally, the

³ <https://hub.worldpop.org/>

⁴ https://www.gebco.net/data_and_products/gridded_bathymetry_data/

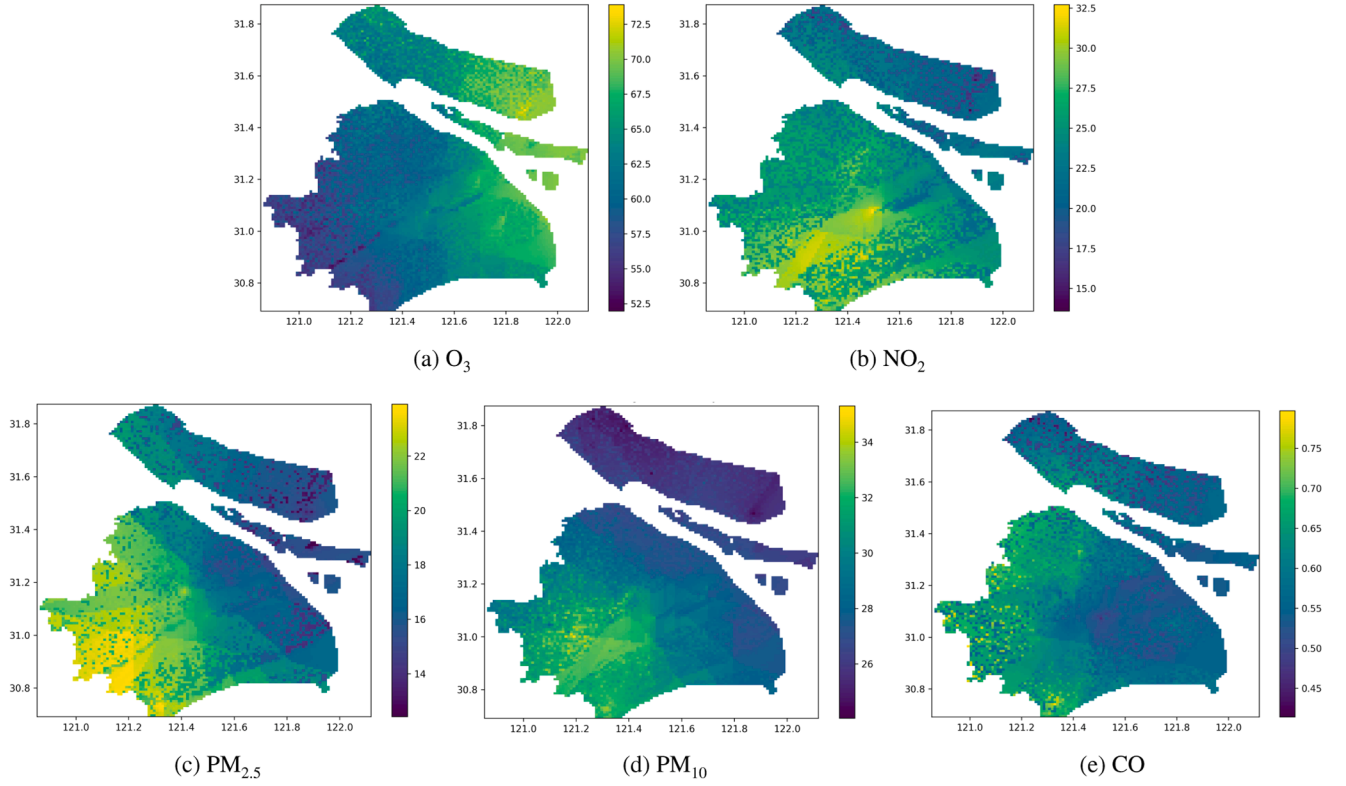


Fig. 7. Visualization of the concentrations of five air quality metrics (O_3 , NO_2 , $PM_{2.5}$, PM_{10} , CO) in Shanghai using the proposed SAQIN model.

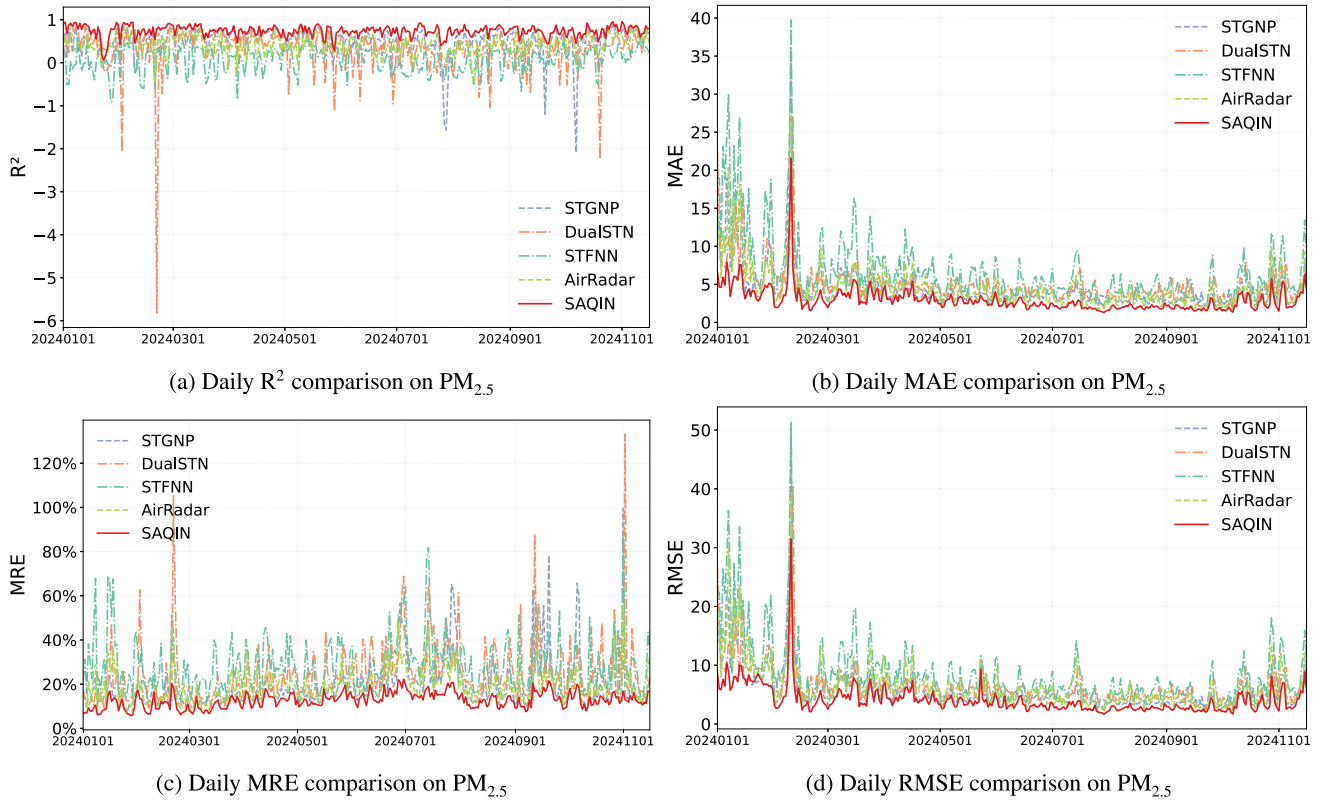


Fig. 8. Daily data reconstruction performance (R^2 , MAE, MRE, RMSE) of deep spatio-temporal methods on $PM_{2.5}$ dataset from 171 air quality monitoring stations in the Yangtze River Delta region during the period from January 1, to November 16, 2024.

Table 4
Performance comparison of different models on air quality metrics (O₃, NO₂, PM_{2.5}, PM₁₀, CO) in Yangtze River Delta region.

Method	Year	O ₃			NO ₂			PM _{2.5}			PM ₁₀			CO			
		R ² ↑	MAE↓	MRE↓	RMSE↓	R ² ↑	MAE↓	MRE↓	RMSE↓	R ² ↑	MAE↓	MRE↓	RMSE↓	R ² ↑	MAE↓	MRE↓	RMSE↓
KNN	1967	0.6111	13.15	0.217	17.57	0.5378	7.12	0.389	9.56	0.6989	8.16	0.373	12.36	0.6585	12.55	0.314	18.61
GPR	1996(NIPS)	0.8201	8.51	0.130	11.82	0.6558	6.36	0.358	8.02	0.8475	5.55	0.256	8.13	0.8531	8.05	0.209	11.21
IGNNK	2021(AAAI)	0.8626	7.93	0.127	10.44	0.7370	5.24	0.261	7.21	0.8822	5.17	0.239	7.73	0.8757	7.76	0.201	11.22
INCREASE	2023(WWW)	0.9174	5.80	0.091	8.10	0.7822	4.60	0.242	6.56	0.9283	3.76	0.165	6.03	0.9260	5.73	0.141	8.66
STGNP	2023(SIGKDD)	0.8398	8.50	0.179	11.26	0.7713	5.07	0.351	6.68	0.8948	4.91	0.220	8.28	0.9214	6.02	0.158	9.11
DualSTN	2024(TNNLS)	0.8946	6.90	0.115	9.131	0.8010	4.53	0.243	6.29	0.9002	4.99	0.243	7.34	0.8695	8.50	0.225	11.76
STFNN	2024(IJCAI)	0.7357	10.94	0.181	14.27	0.7494	5.12	0.278	6.84	0.7950	6.48	0.276	9.64	0.7382	10.46	0.254	16.08
AirRadar	2025(AAAI)	0.8618	6.92	0.112	10.47	0.8244	3.80	0.195	5.89	0.8864	4.49	0.190	7.59	0.8846	6.27	0.149	10.81
SAQIN	2025	0.9474	4.70	0.074	6.46	0.8842	3.46	0.180	4.79	0.9542	2.95	0.127	4.82	0.9509	4.64	0.113	7.05
															0.6485	0.09	0.166
															0.13		

Table 5
Cross-domain performance on air quality metrics (O₃, NO₂, PM_{2.5}, PM₁₀, CO). Deep learning-based models are trained on Beijing regional data and evaluated in the Yangtze River Delta.

Method	O ₃			NO ₂			PM _{2.5}			PM ₁₀			CO			
	R ² ↑	MAE↓	RMSE↓	R ² ↑	MAE↓	RMSE↓	R ² ↑	MAE↓	RMSE↓	R ² ↑	MAE↓	RMSE↓	R ² ↑	MAE↓	RMSE↓	
KNN	0.6111	13.15	0.217	0.5378	7.12	0.389	0.6989	8.16	0.373	12.36	0.6585	12.55	0.314	18.61	0.1695	0.19
GPR	0.8201	8.51	0.130	0.6558	6.36	0.358	0.8475	5.55	0.256	8.13	0.8531	8.05	0.209	11.21	0.3064	0.17
IGNNK	0.6807	12.43	0.208	0.6361	6.21	0.301	0.7689	7.46	0.349	10.83	0.5639	15.41	0.433	21.03	0.2537	0.18
INCREASE	0.8893	6.95	0.106	0.7681	4.82	0.256	0.9267	3.89	0.164	6.10	0.8292	8.02	0.193	13.16	0.4186	0.16
STGNP	0.6757	12.30	0.246	0.6169	6.56	0.374	0.6882	8.16	0.411	12.73	0.7158	13.15	0.428	17.32	0.3506	0.18
DualSTN	0.7016	11.62	0.228	0.7351	5.31	0.280	0.7664	8.16	0.487	11.23	0.5168	16.87	0.513	22.63	0.3121	0.16
STFNN	0.6963	11.89	0.198	0.6518	6.29	0.330	0.7846	7.08	0.313	10.18	0.4705	18.51	0.602	22.93	0.4676	0.16
AirRadar	0.7639	9.61	0.165	0.7078	5.06	0.276	0.7409	7.36	0.353	11.47	0.7980	8.62	0.216	14.31	0.3188	0.17
SAQIN	0.9198	5.74	0.093	0.7927	4.54	0.241	0.9387	3.44	0.149	5.58	0.9422	5.02	0.121	7.66	0.4980	0.15

Table 6

Ablation study of different components in the proposed air quality inference model on the NO₂ dataset (Yangtze River Delta), the O₃ dataset (Shenzhen), as well as the computational efficiency metrics (parameters, FLOPs, and inference time).

Configuration	NO ₂ (Yangtze River Delta)				O ₃ (Shenzhen)				Efficiency		
	R ² ↑	MAE↓	MRE↓	RMSE↓	R ² ↑	MAE↓	MRE↓	RMSE↓	Params (M)↓	FLOPs (G)↓	Time (ms)↓
w/o Semantic Seg	0.8806	3.5350	0.1872	4.8602	0.9303	5.4595	0.1190	7.4071	2.4193	0.0982	66.79
w/o Lightweight Global	0.8823	3.4996	0.1848	4.8262	0.9316	5.4550	0.1141	7.3391	39.2920	2.2000	96.56
w/o RS Image Features	0.8722	3.6458	0.1952	5.0278	0.9275	5.6025	0.1163	7.5534	0.1028	0.0004	51.52
w/o Geospatial variables	0.8821	3.4939	0.1826	4.8300	0.9313	5.3810	0.1176	7.3536	41.4546	2.2973	110.33
w/o Geo Relationship	0.8816	3.5197	0.1851	4.8397	0.9298	5.4633	0.1166	7.4340	41.4592	2.2973	108.34
w/o Multi-pollutant	0.8810	3.5019	0.1843	4.8511	0.9289	5.4975	0.1167	7.4814	41.4576	2.2973	109.55
w/o Residual Correction	0.8767	3.5833	0.1929	4.9392	0.9282	5.5431	0.1184	7.5165	41.4604	2.2973	108.46
Full model	0.8842	3.4645	0.1802	4.7860	0.9334	5.3270	0.1126	7.2385	41.4604	2.2973	113.09

Table 7

Ablation study of the proposed loss function and the individual loss components ($\mathcal{L}_{\text{smooth}}$, $\mathcal{L}_{\text{log}}$, and $\mathcal{L}_{\text{scale}}$) on the O₃ dataset (Yangtze River Delta) and PM_{2.5} dataset (Shenzhen), demonstrating the effectiveness of jointly optimizing for spatial smoothness, relative error robustness, and concentration scale consistency.

Loss Configuration	O ₃ (Yangtze River Delta)				PM _{2.5} (Shenzhen)			
	R ² ↑	MAE↓	MRE↓	RMSE↓	R ² ↑	MAE↓	MRE↓	RMSE↓
$\mathcal{L}_{\text{smooth}}$	0.9460	4.7533	0.0754	6.5474	0.9291	1.9040	0.1420	2.8290
$\mathcal{L}_{\text{log}}$	0.9449	4.7882	0.0759	6.6135	0.9275	1.9211	0.1406	2.8610
$\mathcal{L}_{\text{scale}}$	0.9455	4.7780	0.0755	6.5801	0.9273	1.9273	0.1404	2.8637
$\mathcal{L}_{\text{smooth}} + \mathcal{L}_{\text{log}}$	0.9461	4.7377	0.0749	6.5415	0.9293	1.8995	0.1405	2.8240
$\mathcal{L}_{\text{smooth}} + \mathcal{L}_{\text{scale}}$	0.9463	4.7404	0.0754	6.5292	0.9293	1.9032	0.1407	2.8239
$\mathcal{L}_{\text{log}} + \mathcal{L}_{\text{scale}}$	0.9445	4.8056	0.0754	6.6383	0.9288	1.9051	0.1414	2.8343
$\mathcal{L}_{\text{smooth}} + \mathcal{L}_{\text{log}} + \mathcal{L}_{\text{scale}}$	0.9474	4.6993	0.0744	6.4590	0.9301	1.8817	0.1391	2.8091

RMSE comparison further confirms the superiority of SAQIN by demonstrating a lower overall variance in the prediction errors. These results demonstrate that SAQIN is more accurate and stable than any of the baseline methods evaluated in capturing the complex spatio-temporal dynamics of PM_{2.5}.

4.7. Cross-domain evaluation

To assess the domain generalization capability of SAQIN, we train the model on air quality and environmental data from the Beijing region, which includes 32 monitoring stations and spans from January 2021 to December 2023. The trained model is then directly applied to the Yangtze River Delta region in 2024 without any fine-tuning or adaptation. This zero-shot transfer setting is challenging because the two regions differ substantially in geographic scale, urban morphology, emission composition, and pollution regimes. Despite these distributional shifts, SAQIN achieves strong performance across all five pollutants, as shown in Table 5. In particular, it attains R² scores of 0.9198 for O₃, 0.9387 for PM_{2.5}, and 0.9422 for PM₁₀, significantly outperforming both classical and deep learning baselines.

This robustness stems from SAQIN's residual correction paradigm. By predicting concentration offsets relative to locally observed reference stations, the model leverages the reference concentration C_i , which already embeds the local static baseline, as an anchor, and only learns to map local context to the residual perturbation field. Since this residual field is governed by universal atmospheric processes such as diffusion and advection, the learned mapping generalizes well across regions with distinct emission structures and urban forms. This mechanism enables reliable inference without requiring knowledge of absolute regional concentration levels.

4.8. Ablation and hyperparameter analysis

To verify that the proposed SAQIN is consistent with the physical mechanisms of air pollution and robust across different regions, we perform a set of experiments on the primary dataset in Yangtze River Delta region and a dataset in Shenzhen. Specifically, the NO₂ dataset (Yangtze

River Delta) and O₃ dataset (Shenzhen) are used to evaluate the proposed modules, as summarized in Table 6. Meanwhile, the O₃ dataset (Yangtze River Delta) and PM_{2.5} dataset (Shenzhen) are employed to verify the designed loss function, which are shown in Table 7.

To systematically evaluate the contribution of each component, we constructed several ablation variants corresponding to the configurations in Table 6. Furthermore, to provide a comprehensive evaluation, the computational efficiency metrics (i.e., Parameters, Floating Point Operations (FLOPs), and Inference Time) for each model configuration are also reported. Regarding static features, **w/o Semantic Seg** and **w/o Lightweight Global** remove the semantic segmentation extractor and RepViT-based global extractor, respectively, while **w/o RS Image Features** excludes the entire remote sensing branch. **w/o Geospatial variables** removes population density and elevation inputs. Regarding dynamic features, **w/o Geo Relationship** excludes distance and bearing embeddings, while **w/o Multi-pollutant** ignores chemical coupling by using only the target pollutant. Finally, **w/o Residual Correction** replaces the residual learning module with direct regression, and **Full model** denotes the complete SAQIN framework.

4.8.1. Static feature analysis

Air quality is influenced by both persistent static factors and transient dynamic processes. SAQIN separates these two aspects through dedicated feature branches. As shown in Table 6, removing the semantic segmentation branch or the lightweight global feature branch leads to performance degradation, confirming that fine-grained land-use patterns and coarse environmental priors play important roles in the air quality inference. Excluding image-based dynamic features has the worst performance, highlighting their importance in capturing spatio-temporal variations of pollution. Beyond predictive accuracy, these visual branches also dominate the computational cost. As indicated in Table 6, the static feature extraction module accounts for the vast majority of the parameters, primarily due to the semantic segmentation backbone. In contrast, in the absence of image-based static features, the parameter count for the core spatio-temporal modeling drops to 0.10M with less than 0.01 G FLOPs. It is worth noting that while the theoretical FLOPs decrease by orders of magnitude, the empiri-

Table 8

Performance and efficiency comparison of different global feature extractors on the Yangtze River Delta NO₂ dataset. FLOPs and parameters are calculated for the specific module only.

Model Variant	R ² ↑	MAE↓	MRE↓	RMSE↓	Params (M)	FLOPs (G)
ResNet50	0.8839	3.4649	0.1813	4.7922	23.77	4.13
Standard CNN + SE Block	0.8828	3.4882	0.1842	4.8159	0.11	0.52
Ours (RepViT)	0.8842	3.4645	0.1802	4.7860	2.21	0.40

Table 9

Performance comparison of different structures for the reference dynamic feature extractor on the Yangtze river delta NO₂ dataset.

Structure Variant	R ² ↑	MAE↓	MRE↓	RMSE↓
Attention ×1 (Default)	0.8842	3.4645	0.1802	4.7860
Attention ×2	0.8843	3.4661	0.1802	4.7836
Attention ×3	0.8844	3.4582	0.1782	4.7827
RepViT	0.8832	3.4947	0.1896	4.8071

cal inference time only reduces by roughly half (from 113.09 ms to 51.52 ms). This non-linear scaling occurs because the latency of the core spatio-temporal module is predominantly governed by memory-bound operations, such as geometric distance derivations and topological feature aggregations, rather than the purely parameterized computations measured by FLOPs. Nevertheless, this explicitly demonstrates that the inherent spatio-temporal inference algorithm is exceptionally lightweight. While the visual backbone introduces a heavier parameter burden, it is a worthwhile trade-off for capturing high-resolution spatial non-stationarity. Finally, the overall inference time of 113.09 ms remains highly efficient for real time deployment.

We also examine the contributions of structured geospatial variables such as population density and elevation, which lead to noticeable decrease in accuracy in Table 6, showing that these variables encode anthropogenic emission intensity and terrain-induced accumulation effects that cannot be fully captured by imagery alone.

To further justify the architectural choice of RepViT for the global branch, we conducted comparative studies against a standard heavy-weight backbone (ResNet50) and a lightweight attention baseline (Standard CNN + SE Block). Specifically, this lightweight baseline consists of three sequential Conv-BN-ReLU downsampling stages (with 32, 64, and 128 channels, respectively), incorporating SE Block in the latter two stages, followed by global average pooling and a linear projection. The results on the Yangtze River Delta NO₂ dataset are presented in Table 8. Although ResNet50 achieves competitive accuracy, it incurs significantly higher computational costs (23.77 M parameters and 4.13 G FLOPs). The SE Block variant has much fewer parameters, which yields inferior accuracy (R² = 0.8828) and higher FLOPs compared to RepViT. In contrast, our RepViT-based design achieves the highest R² (0.8842) with the lowest FLOPs (0.40 G), demonstrating an optimal balance between representation capability and computational efficiency.

4.8.2. Dynamic feature analysis

Explicit geometric embeddings (distance and bearing) are crucial for modeling spatial decay and directional transport. As shown in Table 6, removing these features results in consistent performance degradation (e.g., R² drops from 0.8842 to 0.8816 on Yangtze River Delta, and to 0.9298 on Shenzhen). This confirms that the model relies on relative spatial coordinates to learn anisotropic dispersion patterns and assign appropriate attention weights.

Atmospheric pollutants interact through complex chemical and physical processes. According to Table 6, when SAQIN is limited to a single-pollutant input, the model's R² drops from 0.8842 to 0.8810 on the Yangtze River Delta dataset. A consistent decline is observed on the Shenzhen dataset, where the R² decreases from 0.9334 to 0.9289, confirming that jointly modeling multiple pollutant species enhances inference fidelity and stability.

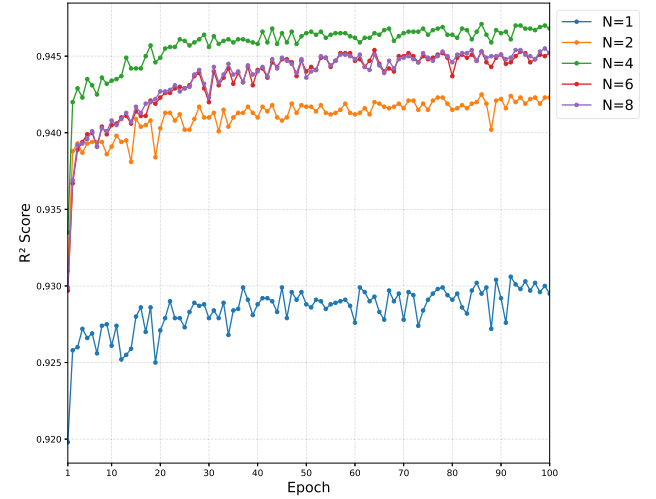


Fig. 9. R² results of the first 100 epochs with different values of N , where it can be observed that the performance is optimal when $N = 4$.

To demonstrate the proposed Reference Dynamic Feature Extractor, we compared the default single-layer multi-head self-attention with deeper variations and a lightweight CNN-based alternative (RepViT). The results are shown in Table 9. Increasing the depth of the attention mechanism yields negligible performance gains (R² improves to 0.8844) but increases computational cost. This observation suggests that the essential spatio-chemical dependencies are adequately captured by a single attention layer, and further stacking layers lead to diminishing returns on performance. Furthermore, replacing multi-head self-attention with RepViT leads to a performance deterioration (R² decreases to 0.8832), as the permutation-invariant self-attention mechanism is more suitable for processing unordered station sets than grid-based CNNs.

4.8.3. Residual correction

To assess the contribution of residual correction, we replace it with a direct concentration regression head. As indicated in Table 6, this variant achieves an R² of 0.8767 on the Yangtze River Delta dataset, which is 0.0075 lower than the full model. Similarly, on the Shenzhen dataset, removing this module leads to a degradation in R² from 0.9334 to 0.9282. These results demonstrate that residual correction is essential for achieving high-fidelity inference. Regarding computational overhead, the results in Table 6 reveal that the residual correction module maintains the same parameter count and FLOPs as the full model at the reported precision. The marginal increase in inference time from 108.46 ms to 113.09 ms represents a minimal cost for the resulting gains in predictive performance, further validating the efficiency of our residual learning design.

4.8.4. Multi-station collaboration

SAQIN estimates target air quality by integrating information from multiple reference stations. This process depends on two key factors: the inclusion of geographical information and the number of reference stations N . Removing geographical information from reference station representations causes a clear performance degradation in Table 6, demonstrating that spatial context is crucial for cross-location generalization.

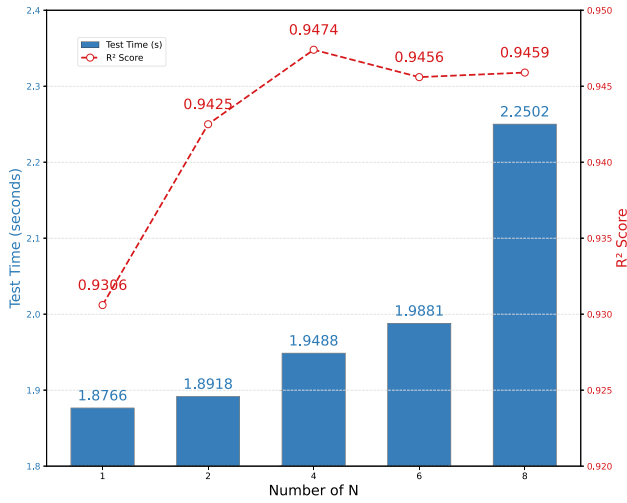


Fig. 10. Comparison of computational time and R^2 for testing 171 stations within Yangtze River Delta region during the testing period with different N .

Table 10

Computational complexity comparison between SAQIN and state-of-the-art models.

Model	Params (M)↓	FLOPs (G)↓	Time (ms)↓
IGNNK	0.0732	0.0293	3.21
INCREASE	0.2161	0.0992	46.97
STGNP	0.1079	1.0522	1096.31
DualSTN	0.0055	0.0055	59.53
STFNN	0.0207	0.4188	88.07
AirRadar	0.1677	0.0521	29.72
SAQIN (w/o RS Image Features)	0.1028	0.0004	51.52
SAQIN	41.4604	2.2973	113.09

The parameter N controls the trade-off between representational diversity and computational cost. As illustrated in Fig. 9, the R^2 reaches its peak at $N = 4$. Fig. 10 further shows that inference time grows slowly for $N \leq 6$ but increases sharply at $N = 8$ with minimal accuracy improvement. Therefore, $N = 4$ provides the best balance between efficiency and performance.

4.8.5. Loss function design

To validate the effectiveness of the composite loss, we evaluate all individual and combined configurations on the O_3 dataset. As shown in Table 7, the complete formulation $\mathcal{L}_{\text{smooth}} + \mathcal{L}_{\log} + \mathcal{L}_{\text{scale}}$ achieves the highest R^2 and the lowest MAE, MRE, and RMSE.

These findings confirm that jointly optimizing for smoothness ($\mathcal{L}_{\text{smooth}}$), logarithmic robustness ($\mathcal{L}_{\log}$), and scale consistency ($\mathcal{L}_{\text{scale}}$) is crucial for reliable air quality inference. While any pairwise combination improves over single losses, only the full composite captures the full spectrum of error characteristics in real-world monitoring data.

4.9. Computational complexity analysis

We further evaluate the computational complexity of the proposed SAQIN framework to assess its feasibility for real-time deployment. The evaluation is conducted on an NVIDIA RTX 3090 GPU. To provide a comprehensive assessment, Table 10 compares our model against several state-of-the-art baselines. In this comparison, we report the metrics for two configurations: the complete multi-source framework (SAQIN) and a streamlined variant denoted as SAQIN (w/o RS Image Features). As detailed in our earlier ablation study (Table 6), this variant isolates our core spatio-temporal inference logic by excluding high-resolution visual priors. Importantly, while the lack of these visual priors leads to a slight performance degradation compared to the complete model,

this streamlined variant still shows robust inference accuracy, which is highly competitive with other state-of-the-art baselines.

In terms of theoretical complexity, the core spatio-temporal inference logic of SAQIN is exceptionally lightweight. As indicated in Table 10, the SAQIN (w/o RS Image Features) variant requires only 0.1028 M parameters and 0.0004 G FLOPs, making it highly competitive with existing signal-based models such as AirRadar (0.1677 M) and INCREASE (0.2161 M). Integrating the high-resolution remote sensing feature branch in the complete SAQIN model introduces a heavier parameter burden (41.4604 M) and higher FLOPs (2.2973 G). However, the time only increases from 51.52 ms to 113.09 ms, which is dedicated to capture complex spatial non-stationarity for performance improvement.

Beyond theoretical metrics, we also evaluate the empirical inference time, specifically measuring the latency to simultaneously infer all 171 target stations in the Yangtze River Delta dataset at a batch size of 32. This empirical speed is often predominantly governed by memory-bound spatial distance derivations rather than purely parameterized operations. This distinction is particularly evident for STGNP. Despite its modest parameter count, the neural process formulation requires the explicit computation of intricate relationships, such as spatial distances and attention weights, between each target node and all available context stations. This inherently sequential one-to-many matching and aggregation process fundamentally prohibits efficient GPU parallelization, resulting in an exceptionally high latency of 1096.31 ms. In contrast, our core variant processes all spatial nodes concurrently through holistic topological operations, achieving a competitive latency of 51.52 ms. For the complete SAQIN model, the inference requires 113.09 ms. Although this multi-source architecture is inherently more complex than pure signal-based models like IGNNK (3.21 ms latency), it is a rewarding trade-off for significant inference performance gains. Considering that air quality monitoring typically operates on an hourly timescale, this inference time is sufficient for high-resolution real time deployment.

5. Conclusion

In this paper, we propose a static-dynamic two-branch SAQIN framework for high-resolution air quality inference by formulating the problem as residual correction task relative to reference stations instead of direct concentration regression problem. Static priors are derived from semantic segmentation and geospatial context, while dynamic features are captured through multi-pollutant interactions and spatial relationships. Residual learning module based on multiple stations are designed to eliminate unknown spatial bias, which is then fused using distance-weighted averaging to remove discontinuities caused by Voronoi partitioning and to ensure global continuity in the reconstructed field. Based on the numerical evaluation over the Yangtze River Delta dataset, SAQIN consistently outperforms state-of-the-art methods and demonstrates robust cross-domain generalization performance. One limitation of this paper lies in its reliance on geometric proxies to represent atmospheric transport, which may not fully capture complex meteorological dynamics. The proposed SAQIN framework distinguishes between static priors and dynamic features, achieving air quality inference through residual learning. This framework provides a unified, physically grounded approach to reconstructing environmental data in regions where information is scarce.

CRediT authorship contribution statement

Hui Zheng: Writing – original draft, Methodology; **Jialing Huang:** Methodology; **Huiming Chen:** Methodology; **Zhenlong Xiao:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition; **Nengwang Chen:** Supervision; **Xinghao Ding:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was supported by the National Key R&D Program of China under grant 2023YFC3804904, and the National Natural Science Foundation of China under Grant 62271430.

References

- Arya, S. P. et al. (1999). *Air pollution meteorology and dispersion* (vol. 310). Oxford University Press, New York.
- Baran, B. (2021). Air quality index prediction in besiktas district by artificial neural networks and k nearest neighbors. *Mühendislik Bilimleri ve Tasarım Dergisi*, 9(1), 52–63.
- Bezyk, Y., Sówka, I., Górka, M., & Blachowski, J. (2021). Gis-based approach to spatio-temporal interpolation of atmospheric Co₂ concentrations in limited monitoring dataset. *Atmosphere*, 12(3), 384.
- Bonet, E. R., Do, T. H., Qin, X., Hofman, J., La Manna, V. P., Philips, W., & Deligiannis, N. (2022). Explaining graph neural networks with topology-aware node selection: Application in air quality inference. *IEEE Transactions on Signal and Information Processing over Networks*, 8, 499–513.
- Chen, L., Papandreou, G., Schroff, F., & Adam, H. (2017). Rethinking atrous convolution for semantic image segmentation. arXiv:1706.05587
- Chen, Y., Liu, S., & Wang, X. (2021). Learning continuous image representation with local implicit image function. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 8628–8638).
- Cheng, W., Shen, Y., Zhu, Y., & Huang, L. (2018). A neural attention model for urban air quality inference: Learning the weights of monitoring stations. In *Proceedings of the AAAI conference on artificial intelligence* (vol. 32).
- Cheng, Y., Zhang, W., Zhang, Z., Kang, J., Xu, Q., Wang, S., & Niyato, D. (2025). SnapCFL: A pre-clustering-based clustered federated learning framework for data and system heterogeneities. *IEEE Transactions on Mobile Computing*, 24(6), (pp. 5214–5228).
- Cressie, N. (2015). *Statistics for spatial data*. John Wiley & Sons.
- Fadhel, M. A., Duhaim, A. M., Saihood, A., Sewify, A., Al-Hamadani, M. N. A., Albahri, A. S., Alzubaidi, L., Gupta, A., Mirjalili, S., & Gu, Y. (2024). Comprehensive systematic review of information fusion methods in smart cities and urban environments. *Information Fusion*, 107, (p. 102317).
- Feng, Y., Wang, Q., Xia, Y., Huang, J., Zhong, S., & Liang, Y. (2024). Spatio-temporal field neural networks for air quality inference. In *Proceedings of the thirty-third international joint conference on artificial intelligence, IJCAI-24* (pp. 7260–7268).
- Guo, G., Wang, H., Bell, D., Bi, Y., & Greer, K. (2003). Knn model-based approach in classification. In *On the move to meaningful internet systems 2003: CoopIS, DOA, and ODBASE* (pp. 986–996). Springer.
- Han, J., Liu, H., Zhu, H., & Xiong, H. (2023). Kill two birds with one stone: A multi-view multi-adversarial learning approach for joint air quality and weather prediction. *IEEE Transactions on Knowledge and Data Engineering*, 35(11), 11515–11528.
- Han, Q., Lu, D., Chen, R. et al. (2021). Fine-grained air quality inference via multi-channel attention model. In *IJCAI* (pp. 2512–2518).
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770–778).
- Hoek, G., Beelen, R., De Hoogh, K., Vienneau, D., Gulliver, J., Fischer, P., & Briggs, D. (2008). A review of land-use regression models to assess spatial variation of outdoor air pollution. *Atmospheric Environment*, 42(33), 7561–7578.
- Hsieh, H., Lin, S., & Zheng, Y. (2015). Inferring air quality for station location recommendation based on urban big data. In *Proceedings of the 21th ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 437–446).
- Hu, J., Liang, Y., Fan, Z., Chen, H., Zheng, Y., & Zimmermann, R. (2023a). Graph neural processes for spatio-temporal extrapolation. In *Proceedings of the 29th ACM SIGKDD conference on knowledge discovery and data mining* (pp. 752–763).
- Hu, J., Liang, Y., Fan, Z., Liu, L., Yin, Y., & Zimmermann, R. (2023b). Decoupling long-and short-term patterns in spatiotemporal inference. *IEEE Transactions on Neural Networks and Learning Systems*, 35(11), 16328–16340.
- Huang, J., Duan, N., Ji, P., Ma, C., Ding, Y., Yu, Y., Zhou, Q., Sun, W. et al. (2018). A crowdsourcing-based sensing system for monitoring fine-grained air quality in urban environments. *IEEE Internet of Things Journal*, 6(2), 3240–3247.
- Huang, W., Li, T., Liu, J., Xie, P., Du, S., & Teng, F. (2021). An overview of air quality analysis by big data techniques: Monitoring, forecasting, and traceability. *Information Fusion*, 75, 28–40.
- Jerrett, M., Arain, A., Kanaroglou, P., Beckerman, B., Potoglou, D., Sahuvaroglu, T., Morrison, J., & Giovis, C. (2005). A review and evaluation of intraurban air pollution exposure models. *Journal of Exposure Science & Environmental Epidemiology*, 15(2), 185–204.
- John, A., Redmond, S. J., Cardiff, B., & John, D. (2021). A multimodal data fusion technique for heartbeat detection in wearable IoT sensors. *IEEE Internet of Things Journal*, 9(3), 2071–2082.
- Jutzeler, A., Li, J., & Faltings, B. (2014). A region-based model for estimating urban air pollution. In *Proceedings of the AAAI conference on artificial intelligence*. (vol. 28).
- Karagulian, F., Belis, C. A., Dora, C. F. C., Prüss-Ustün, A. M., Bonjour, S., Adair-Rohani, H., & Amann, M. (2015). Contributions to cities' ambient particulate matter (PM): A systematic review of local source contributions at global level. *Atmospheric Environment*, 120, 475–483.
- Li, J., & Heap, A. D. (2011). A review of comparative studies of spatial interpolation methods in environmental sciences: Performance and impact factors. *Ecological Informatics*, 6(3–4), 228–241.
- Lin, Y., Chiang, Y.-Y., Pan, F., Stripelis, D., Ambite, J. L., Eckel, S. P., & Habre, R. (2017). Mining public datasets for modeling intra-city PM_{2.5} concentrations at a fine spatial resolution. In *Proceedings of the 25th ACM SIGSPATIAL international conference on advances in geographic information systems* (pp. 1–10).
- Liu, J., Li, T., Xie, P., Du, S., Teng, F., & Yang, X. (2020). Urban big data fusion based on deep learning: An overview. *Information Fusion*, 53, 123–133.
- Loshchilov, I. (2017). Decoupled weight decay regularization. arXiv:1711.05101
- Loshchilov, I., & Hutter, F. (2016). SGDR: Stochastic gradient descent with warm restarts. arXiv:1608.03983
- Ma, R., Liu, N., Xu, X., Wang, Y., Noh, H. Y., Zhang, P., & Zhang, L. (2020). Fine-grained air pollution inference with mobile sensing systems: A weather-related deep autoencoder model. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 4(2), 1–21.
- Marchang, N., & Tripathi, R. (2020). KNN-ST: Exploiting spatio-temporal correlation for missing data inference in environmental crowd sensing. *IEEE Sensors Journal*, 21(3), 3429–3436.
- Miller, H. J. (2004). Tobler's first law and spatial analysis. *Annals of the Association of American Geographers*, 94(2), 284–289.
- Organization, W. H. et al. (2015). Economic cost of the health impact of air pollution in Europe: Clean air, health and wealth. In *Economic cost of the health impact of air pollution in Europe: Clean air, health and wealth*.
- Pasquill, F. (1996). *Atmospheric diffusion: The dispersion of windborne material from industrial and other sources*. Ellis Horwood Ltd.
- Pasquill, F., & Smith, F. B. (1983). *Atmospheric diffusion* (vol. 437). E. Horwood, New York, NY, USA.
- Patel, Z. B., Purohit, P., Patel, H. M., Sahni, S., & Batra, N. (2022). Accurate and scalable Gaussian processes for fine-grained air quality inference. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 12080–12088). (vol. 36).
- Qian, Y., Zhou, W., & Wang, C. (2018). Research on multi-source data fusion in the field of atmospheric environmental monitoring. In *2018 13th international conference on computer science & education (ICCSE)* (pp. 1–5). IEEE.
- Ren, S., He, K., Girshick, R., & Sun, J. (2017). Faster R-CNN: Towards real-time object detection with region proposal networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 39(6), 1137–1149.
- Rowley, A., & Karakus, O. (2023). Predicting air quality via multimodal AI and satellite imagery. *Remote Sensing of Environment*, 293, 113609.
- Seinfeld, J. H., & Pandis, S. N. (2016). *Atmospheric chemistry and physics: From air pollution to climate change*. John Wiley & Sons.
- Shang, J., Zheng, Y., Tong, W., Chang, E., & Yu, Y. (2014). Inferring gas consumption and pollution emission of vehicles throughout a city. In *Proceedings of the 20th ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 1027–1036).
- Song, J., & Stettler, M. E. J. (2022). A novel multi-pollutant space-time learning network for air pollution inference. *Science of The Total Environment*, 811, 152254.
- Su, H., Chen, Y., Tan, H., Wilson, J. P., Bao, L., Chen, R., & Luo, J. (2024). An improved geographic pattern based residual neural network model for estimating PM_{2.5} concentrations. *International Journal of Applied Earth Observation and Geoinformation*, 134, 104174.
- Sun, B., Li, Y., Zhang, Y., & Guo, T. (2024). Multi-source heterogeneous data fusion prediction technique for the utility tunnel fire detection. *Reliability Engineering & System Safety*, 248, 110154.
- Sun, J., Peng, M., Jiang, H., Hong, Q., & Sun, Y. (2022). Hmian: A hierarchical mapping and interactive attention data fusion network for traffic forecasting. *IEEE Internet of Things Journal*, 9(24), 25685–25697.
- Tobler, W. R. (1970). A computer movie simulating urban growth in the detroit region. *Economic Geography*, 46(sup1), 234–240.
- Wang, A., Chen, H., Lin, Z., Han, J., & Ding, G. (2024a). Repvit: Revisiting mobile cnn from vit perspective. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 15909–15920).
- Wang, H., Wang, B., Alharbi, A. M., Gao, D. W., Ma, H., & Luo, P. (2024b). Correlation-based multimodal fusion method for icing degree monitoring of transmission lines within internet of things. *IEEE Internet of Things Journal*, 11(14), 25413–25424.
- Wang, J., Zheng, Z., Ma, A., Lu, X., & Zhong, Y. (2021). LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation. In J. Vanschoren, & S. Yeung (Eds.), *Proceedings of the neural information processing systems track on datasets and benchmarks*. (vol. 1).
- Wang, Q., Xia, Y., ZHong, S., Li, W., Wu, Y., Cheng, S., Zhang, J., Zheng, Y., & Liang, Y. (2025). AirRadar: Inferring nationwide air quality in China with deep neural networks. arXiv:2501.13141
- Wang, Y., Liu, T., Lu, Y., Huang, P., Wan, H., & Deng, F. (2024c). Traffic accident risk prediction of tunnel based on multi-source heterogeneous data fusion. *IEEE Access*, 12, 18694–18702.
- Wang, Y., Song, G., Du, L., & Lu, Z. (2019). Real-time estimation of the urban air quality with mobile sensor system. *ACM Transactions on Knowledge Discovery from Data (TKDD)*, 13(5), 11–19.
- Williams, C., & Rasmussen, C. (1995). Gaussian processes for regression. *Proceedings of the 9th International Conference on Neural Information Processing Systems*(pp. 514–520).
- Wong, D. W., Yuan, L., & Perlin, S. A. (2004). Comparison of spatial interpolation methods for the estimation of air quality data. *Journal of Exposure Science & Environmental Epidemiology*, 14(5), 404–415.

- Wu, D., Xiao, T., Liao, X., Luo, J., Wu, C., Zhang, S., Li, Y., & Guo, Y. (2020). When sharing economy meets IoT: Towards fine-grained urban air quality monitoring through mobile crowdsensing on bike-share system. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 4(2), 1–26.
- Wu, Y., Zhuang, D., Labbe, A., & Sun, L. (2021). Inductive graph neural networks for spatiotemporal kriging. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 4478–4485). (vol. 35).
- Xu, Y., & Zhu, Y. (2016). When remote sensing data meet ubiquitous urban data: Fine-grained air quality inference. In *2016 IEEE International conference on big data (big data)* (pp. 1252–1261). IEEE.
- Xu, Y., Zhu, Y., Shen, Y., & Yu, J. (2019). Fine-grained air quality inference with remote sensing data and ubiquitous urban data. *ACM Transactions on Knowledge Discovery from Data (TKDD)*, 13(5), 1–27.
- Yun, L., & Luo, Z. (2022). Multisource information fusion algorithm for personalized tourism destination recommendation. *Mathematical Problems in Engineering*, 2022(1), 3503548.
- Zaid, M., & Basu, D. (2024). Understanding the relationship between land use/land cover changes and air quality: A GIS-based fuzzy inference system approach. *Environmental Monitoring and Assessment*, 196(12), 1160.
- Zhang, L., Xie, Y., Xidao, L., & Zhang, X. (2018). Multi-source heterogeneous data fusion. In *2018 International conference on artificial intelligence and big data (ICAIBD)* (pp. 47–51). IEEE.
- Zhang, Q., Han, Y., Li, V. O. K., & Lam, J. C. K. (2022a). Deep-AIR: A hybrid CNN-LSTM framework for fine-grained air pollution estimation and forecast in metropolitan cities. *IEEE Access*, 10, 55818–55841.
- Zhang, X., Hu, J., Zhou, P., & Wang, G. (2022b). An improved multi-source spatiotemporal data fusion model based on the nearest neighbor grids for PM2.5 concentration interpolation and prediction. In *International conference on data mining and big data* (pp. 273–287). Springer.
- Zhao, X., Xu, T., Fu, Y., Chen, E., & Guo, H. (2017). Incorporating spatio-temporal smoothness for air quality inference. In *2017 IEEE International conference on data mining (ICDM)* (pp. 1177–1182). IEEE.
- Zhao, Y., Tao, Z., Wang, M., Chen, Y., Wu, R., & Guo, L. (2022). Landscape ecological risk assessment and planning enlightenment of songhua river basin based on multi-source heterogeneous data fusion. *Water*, 14(24), 4060.
- Zheng, C., Fan, X., Wang, C., Qi, J., Chen, C., & Chen, L. (2023a). Increase: Inductive graph representation learning for spatio-temporal kriging. In *Proceedings of the ACM web conference 2023* (pp. 673–683).
- Zheng, D., Shen, G., Li, Y., Cao, X., & Mukherjee, B. (2023b). Service function chaining and embedding with heterogeneous faults tolerance in edge networks. *IEEE Transactions on Network and Service Management*, 20(3), 2157–2171.
- Zheng, Y., Liu, F., & Hsieh, H.-P. (2013). U-air: When urban air quality inference meets big data. In *Proceedings of the 19th ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 1436–1444).